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Early PCOS Detection: A Comparative Analysis of Traditional and Ensemble Machine Learning Models With Advanced Feature Selection

Khandaker Mohammad Mohi Uddin¹  | Md. Tofael Ahmed Bhuiyan¹ | Md. Mahbubur Rahman² | Md. Manowarul Islam³  | Md Ashraf Uddin⁴

¹Department of Computer Science and Engineering, Southeast University, Dhaka, Bangladesh | ²Department of Computer Science and Engineering, Bangladesh University of Business and Technology, Dhaka, Bangladesh | ³Department of Computer Science and Engineering, Jagannath University, Dhaka, Bangladesh | ⁴School of Information and Technology, Deakin University, Warun Ponds, Australia

Correspondence: Khandaker Mohammad Mohi Uddin (jilanicsejnu@gmail.com)

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ABSTRACT

PCOS (polycystic ovary syndrome) is a common hormonal disorder that affects many women during their reproductive years. It is marked by hormonal imbalances, leading to ovarian cysts, and can result in health issues such as infertility, diabetes, and even heart problems. Diagnosing PCOS accurately and early can be challenging, as it requires specific medical expertise. However, spotting PCOS promptly allows individuals to follow medical recommendations, which can lead to healthier lifestyles. In this study, we examined a dataset consisting of 541 patient records to enhance the detection of PCOS using advanced machine learning techniques. We established a data preprocessing pipeline that rigorously addressed missing values and identified outliers, while also normalizing the data to ensure it was ready for input. For feature selection, we applied advanced techniques such as SelectKBest, Chi-Square, and XGBoost. These methods helped us pinpoint the most predictive attributes, which improved the interpretability and efficiency of our models. Hyperparameter tuning was carefully performed through grid search and cross-validation, ensuring that each model was optimized for the best prediction accuracy. Importantly, our research highlights how effective machine learning can be in predicting PCOS. The logistic regression and support vector machine model stood out with its remarkable accuracy of 99.7753%. Furthermore, we created a user-friendly web application to facilitate smooth deployment and real-time analysis. This provides healthcare professionals with a handy tool for identifying early risks related to PCOS. The web application features an intuitive interface where users can easily input clinical information and receive immediate risk assessments.

1 | Introduction

PCOS, or polycystic ovary syndrome, is a common endocrinological disorder characterized by an excessive level of the male hormone androgen in females [1, 2]. Multiple ovarian cysts

develop inside the ovaries as a result of this hormonal imbalance, which also impairs normal ovarian function [3]. Despite PCOS being common, its precise etiology remains unclear, making early detection and treatment with efficient diagnostic techniques essential. Medical diagnostics have greatly progressed thanks to

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the powerful capabilities of machine learning in pattern recognition and predictive analytics. The uterus, vagina, fallopian tubes, and ovaries are all parts of the female reproductive system [4].

More importantly, a number of clinical tests performed by qualified medical practitioners are often necessary for an accurate diagnosis of PCOS. This can be difficult in areas with limited resources or a shortage of qualified medical professionals. Due to a lack of access to specialists, many young women, especially those living in rural parts of developing and least developed countries, experience the negative impacts of PCOS. As a result, many cases remain undiagnosed and untreated. To address this issue, various computer-based techniques have been developed to effectively predict PCOS based on test results and symptoms. However, even with their significant accomplishments, conventional machine learning algorithms may struggle to provide accurate outcomes when dealing with large datasets [5].

Globally, 1 in 10 premenopausal women of reproductive age, and 44%–70% of women with reproductive health issues, are affected by PCOS, according to epidemiological research [6]. Serious and potentially life-threatening complications are associated with the condition. Numerous metabolic and psychological issues that lower the quality of life, such as hyperandrogenism, thyroid abnormalities, insulin resistance, irregular periods, anovulation, increased anxiety, and sexual dysfunction, are linked to PCOS [7, 8]. According to research published by Stein and Leventhal in 1935, 5%–10% of women aged 12–45 have PCOS [9]. Over 70% of women worldwide experience it, and it can go untreated for long periods [10]. The prevalence differs among ethnic groups: 4.8% in white women, 8% in African American women, 6.8% in Hispanic women, and 31.3% in Asian women [11]. In India, 15.3% of women suffer from PCOS, significantly impacting their lives and potentially leading to conditions like anxiety, sleep apnea, and metabolic syndromes, which increase the risk of heart disease, diabetes, and endometrial cancer [12]. High levels of testosterone can cause symptoms such as infertility, weight gain, irregular periods, and excessive body hair. Hormonal imbalances disrupt ovulation cycles [13, 14].

Although the precise cause of PCOS is still unknown, insulin resistance is often indicated by patches of darker skin. Lifestyle factors like diet and physical inactivity may also influence the onset of PCOS [15, 16]. The main goal of treatment is to manage symptoms. For example, hormonal birth control can regulate menstrual cycles and reduce testosterone levels, while anti-androgens can help decrease acne and excessive hair growth. Diabetes treatments like metformin can regulate menstrual cycles and insulin levels, and fertility treatments can trigger ovulation. Laparoscopic ovarian drilling may reduce testosterone production and restore ovulation if medications are ineffective [17].

Despite previous advancements in PCOS prediction, much more has to be discovered. The results of certain studies are not generally applicable because they only use the same types of datasets. People are hard to fully trust since they do not use fresh information to support their opinions. Simple methods are frequently used to choose features from previous research, excluding potential correlations between qualities. Moreover,

most models are not designed for practical applications, rendering them ineffective in actual medical settings. We made an effort to determine this. The extensive dataset we utilized had 541 items. With the use of technologies like SelectKBest, Chi-Square, and XGBoost, we were able to choose features more intelligently. To improve our predictions, we then experimented with other ensemble classifiers. And not only that, for ease of use by physicians and others, we developed a web application. It is our method of ensuring that our work is useful to individuals and not simply for academic publications.

Machine learning algorithms can be employed to identify PCOS early, before the condition fully develops. In this study, we propose an intuitive web-based interface that incorporates machine learning techniques to predict PCOS early. Identifying whether a woman has PCOS involves a binary classification task with a yes or no outcome. Several models are used for this prediction, including Multinomial Naive Bayes (MNB), K-Nearest Neighbors (KNNs), support vector machine (SVM), Gaussian Naive Bayes (GNB), and logistic regression (LR). These models can produce accurate predictions when provided with the right data, and each model generates different results. The proposed system utilizes data preprocessing techniques such as early column removal, null value imputation, and string-to-integer conversion. In machine learning-based classification, feature selection is crucial as it improves model interpretability, reduces dimensionality, and enhances performance. Three feature selection methods, namely, XGBoost, SelectKBest, and Chi-Square, were applied in our proposed research. These methods were chosen for their ability to identify key features necessary for PCOS diagnosis. To further enhance performance, we included a voting classifier, scaled features, and fine-tuned hyperparameters. After significant hyperparameter adjustments, the Random Forest (RF), KNN, SVM, LR, and LGBM classifiers were able to predict outcomes more accurately than the previous model. LR and SVM were the most accurate, achieving an outstanding accuracy rate of 99.7753%. Similarly, our study enhances the reliability of PCOS prediction using cutting-edge machine learning approaches and prediction systems. The main contributions of our study are as follows:

- To improve prediction accuracy, we employed eight machine learning techniques: RF, KNN, SVM, CatBoost, AdaBoost, DT, LR, and LGBM classifier. Among them, the LR and SVM model achieved the highest prediction accuracy at 99.7753%.
- Hyperparameter tuning, feature scaling, and the voting classifier were used to maximize model performance.
- To identify the most accurate features, three feature selection methods XGBoost, SelectKBest, and Chi-Square were applied.
- The web application, with its highly dynamic and user-friendly design, is responsive on both desktop and mobile platforms, ensuring accessibility and ease of use for users with varying technical backgrounds.

This paper is structured as follows: Section 2 reviews related work. Section 3 outlines the main methodology, including the feature selection process and the ensemble machine learning approach. Section 4 presents the results and performance analysis. Finally, Section 5 concludes the research.

2 | Related Work

Nowadays, there is growing interest in using machine learning for medical screening, particularly for predicting PCOS. Several studies have explored various feature selection techniques and machine learning algorithms to develop reliable and accurate models for PCOS diagnosis. As medical and biological databases rapidly expand, machine learning and artificial intelligence approaches are becoming increasingly common for analyzing large and complex datasets [18]. In recent years, many researchers have proposed AI-based algorithms for PCOS detection, utilizing clinical factors and vital signs as primary datasets.

Silva et al. first presented the BorutaShap technique in their study, after which they trained a RF model. The dataset included 72 PCOS patients and 73 healthy women. A total of 58 traits were assessed based on their importance and suitability. The best-performing model, RF, achieved an accuracy rate of 93% according to the research. The algorithms were trained and evaluated on the same dataset, which may have led to overfitting. However, a significant drawback was the lack of external validation. Additionally, the study did not examine the integration of clinical decision support systems, which is essential for real-world practical implementation [19].

Rahman et al. used a dataset of 541 medical records to develop a web-based machine learning approach for PCOS prediction. Several classifiers, including AB, RF, Decision Trees (DT), and LR, were employed in their study. Among all the models, AdaBoost and RF demonstrated the highest accuracy of 94%, showcasing how diagnostic accuracy can be improved using ensemble approaches that combine multiple sources of data to enhance model performance. However, the study's primary focus on a small dataset may limit the generalizability of the results. Furthermore, it is important to note that thorough clinical validation is lacking in the web-based implementation, which is essential for assessing the model's applicability [20].

Khanna et al. used an open-source dataset of 541 patients from Kerala, India, to create a model for predicting PCOS in fertile individuals. A variety of machine learning models, such as Naive Bayes (NB), DT, KNNs, RFs, SVM, AdaBoost, Extra Trees, and Gradient Boost, were employed by the authors. They also recommended using multiple machine learning stacks. Explainable AI approaches were incorporated to enhance the interpretability and reliability of model predictions. The findings showed that multi-stacking machine learning produced the best outcomes, with an accuracy of 98%. However, the use of multiple models on a single dataset may lead to overfitting, which is one of the study's limitations. Additionally, the complexity of the multi-stacking approach may hinder its practical application in clinical settings, where simplicity and interpretability are crucial [21].

Bharati et al. focused on the data-driven diagnosis of PCOS in women using an open-source dataset consisting of 541 participants, 177 of whom were diagnosed with PCOS. The study employed a hybrid RFLR model, along with LR, RF, Gradient Boosting, and a univariate feature selection (UFS) approach. To train and test the models, the dataset was divided using both holdout and cross-validation methods. The hybrid RFLR model

combined with UFS achieved the highest accuracy of 91.01%. Using UFS alone may overlook important variable interactions that are vital for PCOS diagnosis. Furthermore, the study did not account for how differences in data quality may impact model performance [22].

Zigarelli et al. investigated machine learning methods to create a prediction model for PCOS self-diagnosis. They used a publicly accessible Kaggle PCOS dataset that included information on 541 patients from 10 different Keralan hospitals, containing 44 features. The authors utilized the CatBoost classifier and assessed its efficacy using K-fold validation. The results showed an accuracy rate of 82.5% for invasive therapies and 90.1% for non-invasive clinical indications [23].

Danaei Mehr and Polat also utilized multiple classifiers and feature extraction techniques, including Extra Tree, AdaBoost, Bagging Ensemble with RF, and MLP classification models. Their proposed Ensemble RF classifier achieved the highest accuracy of 98.89%. They found that some models performed better with reduced feature subsets. However, the study's limitations included a lack of dataset diversity and not exploring more advanced models [24].

In order to develop machine learning models for PCOS diagnosis prediction, Zad et al. [25] used a sizable dataset of 30,601 electronic health records (EHRs). Their techniques, which included RFs, Gradient-Boosted Trees (GBTs), SVMs, and LR, produced AUCs ranging from 77.4% to 85%, with GBT being the best performer. Nevertheless, the absence of external validation restricts the generalizability of their findings despite the big dataset. Furthermore, the study's dependence on complicated methods like recursive feature reduction and MLP scores adds needless complexity without producing appreciable performance gains. Additionally, the lack of a deploying framework limits its practical application in clinical settings. Additionally, the dataset's limited worldwide applicability due to its restriction to a specific geographic region creates possible biases.

Danaei Mehr and Polat concluded that reducing feature subsets improved the performance of certain models after experimenting with multiple classifiers and feature extraction techniques [24]. Their study was limited by dataset homogeneity and not exploring more sophisticated model architectures. Earlier studies on PCOS prediction, such as those by Bharati et al. [22], Silva et al. [19], and Zigarelli et al. [23], made significant strides despite limitations. These studies may have overlooked critical markers by using limited feature sets or single-model approaches. Additionally, many did not incorporate essential techniques like hyperparameter tuning.

A number of recent research investigated the prediction of PCOS by machine learning. When Silva et al. [19] tested using BorutaShap to select features, their RF model achieved a 93% accuracy. However, they failed to test its functionality on various data sets and did not include any features that would facilitate its practical application by physicians.

On 541 records, Rahman et al. [20] and the others applied models such as AdaBoost and RF. Their accuracy rate was 94%, which is rather good. They focused too little on feature selection, despite

the fact that their ensemble approaches improved the model's performance. Their model may become weaker in other contexts as a result.

Multi-stacking machine learning was used by Khanna et al. [21], who obtained a 98% accuracy rate. Despite its effectiveness, the multi-stacking approach's complexity prevents it from being widely used in clinical settings. An accuracy of 91.01% was attained by Bharati et al. [22] using a hybrid RFLR model with UFS. Nevertheless, simple selection ignores intricate feature relationships. Zigarelli et al. [23] used CatBoost to obtain a 90.1% accuracy, but they did not assess the model's suitability for use in real-time diagnosis systems.

Machine learning techniques for PCOS prediction have been investigated in a number of recent research. A 95.7% accuracy rate was attained using RF, AdaBoost, and Extra Tree; nevertheless, generalizability was constrained by their dependence on a single dataset. RF and Bagging Ensemble were used by Danaei Mehr and Polat [24], who achieved a 98.89% accuracy; nevertheless, more advancements were limited because of the absence of different datasets and sophisticated architectures. With AUCs in the range of 77.4%–85%, Zad et al. [25] used RF, GBTs, SVM, and LR for analyzing 30,601 EHRs. But without any discernible improvements, the lack of outside validation and the dependence on intricate techniques like MLP scores and recursive feature selection introduced needless complexity. Clinical usefulness and worldwide application were further restricted by their absence of a deployment mechanism and regional dataset constraint.

Our method addresses these drawbacks and offers notable improvements. To ensure a comprehensive feature set, we employ three feature selection strategies (SelectKBest, Chi-Square, and XGBoost) to identify 14 key features. Various machine learning models, including RF, KNN, SVM, LR, and LGBM, are used to provide an extensive analysis of performance differences. Model performance is optimized through ensemble approaches, feature scaling, and hyperparameter adjustment. Notably, the LR and SVM model achieves an impressive accuracy of 99.7753%, outperforming previous research. Our study makes a substantial contribution to the field of PCOS prediction by incorporating modern techniques and addressing the limitations of prior research.

3 | Methodology

The dataset obtained from Kaggle underwent careful selection and necessary preparation. We addressed null values, eliminated duplicate and unnecessary columns, and methodically converted all data from object type to numerical format. These actions ensured a solid foundation for further analysis. The dataset is now prepared for use with various machine learning models, and we divided it into separate training and testing subsets. Our methodology included three notable feature selection algorithms: SelectKBest, Chi-Square, and XGBoost. By ranking based on their scores, we selected the top 20 features from each method, resulting in 14 common features. Our goal is to evaluate each model using key performance indicators such as ROC score, F1 score,

accuracy, and precision. Through this comprehensive comparison, we will identify the model that yields the best outcomes. Our decision-making will be guided by the findings, allowing us to select the model with optimal performance. The user interface will seamlessly incorporate this chosen paradigm, ensuring a sophisticated and efficient user experience. The operation of our proposed approach is depicted in Figure 1. Every step of the process, from data preparation to performance analysis, is shown in this figure.

3.1 | Dataset Description

In this part, patient symptoms and PCOS diagnostic findings are used to train the machine learning supervised algorithms. The training dataset was obtained from a Kaggle library of PCOS patient information that is openly accessible [26]. The dataset was carefully reviewed before training to make sure it was understood completely. The main dataset comprises 541 medical histories of female patients, with 45 columns including various clinical data pertaining to anomalies associated with PCOS. Interestingly, there is a section called "PCOS(Y/N)" where the diagnosis findings are shown as "Yes" or "No," indicating whether PCOS is present or not. In the dataset, the letters "0" and "1" stand for "negative" and "no," respectively, and "yes" and "positive" for PCOS diagnosis. Columns include data such as weight, height, blood type, pulse rate, cycle length (days), pregnancy status (yes/no), hair loss (yes/no), acne (yes/no), among others. For this study, the column designated for training purposes has been identified as this particular feature column.

3.2 | Correlation Coefficient

The quantifiable link or reliance between two variables is indicated by correlation, which measures how the value of one variable varies in reaction to changes in the other. A number closer to 1 denotes a stronger association, while a value closer to 0 denotes a weaker relationship. Correlation values have scales between -1 and 1 . The connection between components K and M is as follows, as mentioned in [27]. The correlation equation is shown in Equation (1)

$$\rho_{k,m} = \frac{\text{cov}(K, M)}{\sigma_K \sigma_M} \quad (1)$$

3.3 | Data Preprocessing

To improve the dataset before feeding it into various machine learning models, data preparation is essential. Techniques employed in this process include encoding, removing unnecessary data, managing null values, and deleting extraneous columns. The second dataset contains six columns, while the first has 43. These databases encompass findings from tests performed on 541 patients. After merging and removing duplicates based on four shared characteristics, the dataset consists of 45 columns and 541 patients. Columns labeled "Sl. No.," "The Patient File No.," and an unidentified column were eliminated, as they did not contribute to the results. The final data frame contains 42 columns and 541 rows. Text values in the "AMH (ng/mL)" column were converted to numeric values using the

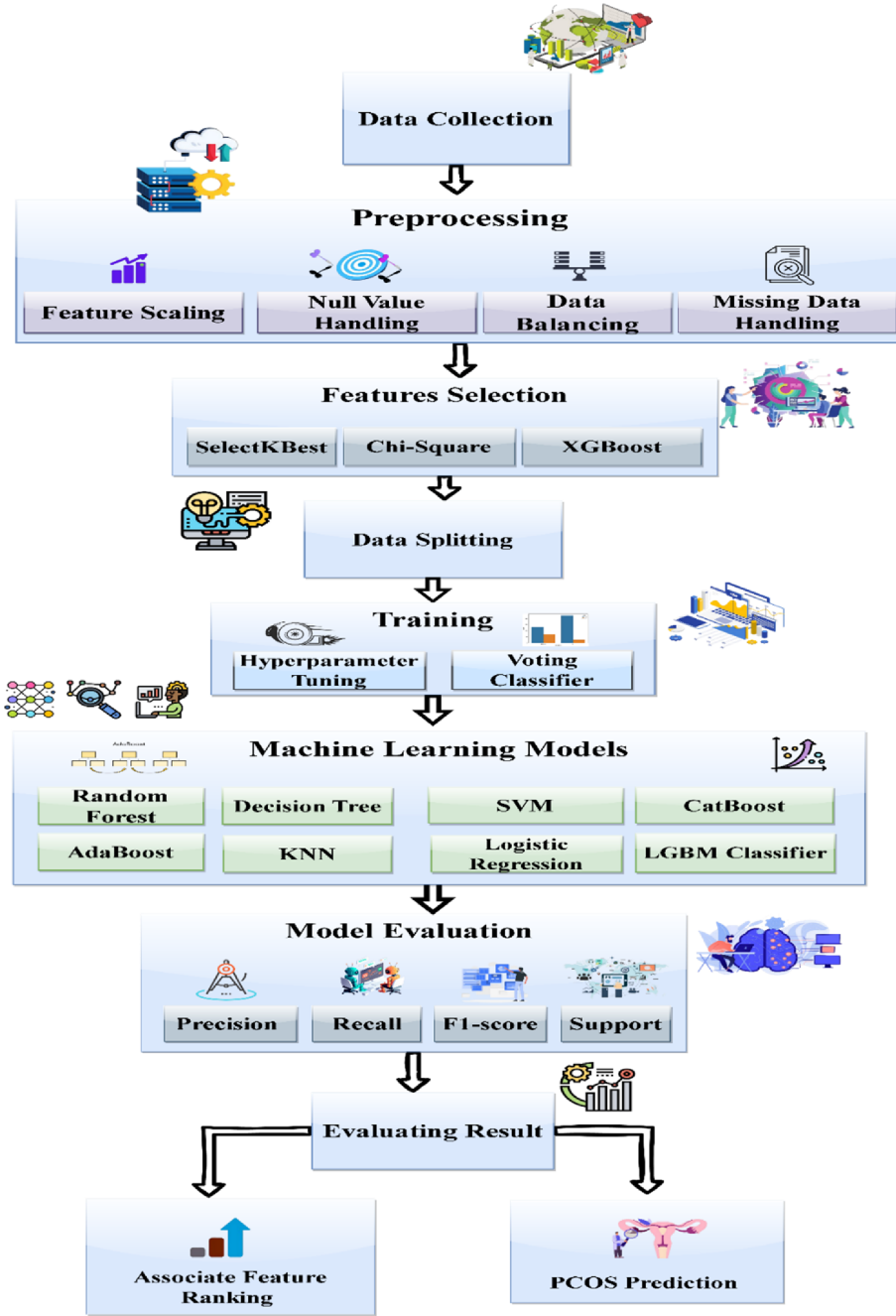


FIGURE 1 | Workflow model for the proposed system.

“to_numeric ()” function from the Pandas library. Additionally, standard deviations for each row were imputed for the null values in the following columns: “Fast Food (Y/N),” “Marriage Status (yrs.),” and “AMH (ng/mL).” The dataset was standardized, excluding the target column, prior to applying machine learning models. Various machine learning techniques were applied during the data preparation and cleaning process [28]. Preprocessing methods addressed missing values, identified outliers, discretized data, and reduced both numerosity and dimensionality to handle inconsistent and incomplete data. Missing values were filled using imputation techniques, while label encoding was utilized to convert categorical data into numerical format. Finally, feature selection was employed to eliminate unnecessary data points.

3.4 | Label Encoding

One popular method for encoding information based on categories is label encoding. Based on the alphabetical order of the categories, it gives each one a distinct numerical value. The following mathematical formulae are used to compute the encoded categories as Equations (2) and (3)

$$S = \frac{1}{1 + \exp\left(\frac{n-mdl}{a}\right)} \quad (2)$$

$$x^k = \text{prior} * (1 - s) + s * \frac{n^+}{n} \quad (3)$$

Here, a denotes the smoothing parameter, while mdl signifies the minimum number of samples in a leaf, influencing the regularization strength. Recommended values for mdl and a fall within the range of 1–100. Unique new values specific to the training dataset are employed to substitute the existing ones [22].

3.5 | Data Analysis

The dataset is validated by extensive data analysis to guarantee correctness. Finding the connections, patterns, and trends necessary for wise decision-making is the aim. There are methodical phases in the process: first dataset collection from a Kaggle dataset that is made accessible to the public. It consists of 541 records of patients with 45 PCOS-related variables, including biochemical and clinical indicators. Data preparation eliminates redundant and unneeded columns, uses imputation techniques to handle null values, and transforms data from the type of object to numerical representation. The dataset is then divided into subgroups for testing and training. Three different algorithms—SelectKBest, Chi-Square, and XGBoost—are used in feature selection to determine the top 20 features from each approach. This process yields the discovery of 14 features that are shared by all three methods. Several machine learning models were developed as part of the model development process, including LGBM Classifier, RF, SVM, KNN, AdaBoost, DT, and LR. Strict hyperparameter tuning, feature scaling, and the use of ensemble methods like stacking and voting classifiers were used to apply optimization strategies. AUC, ROC curve, F1 score, accuracy, precision, recall, and F1 score were among the measures used to evaluate performance. Model performance was demonstrated through the use of visual aids including confusion matrices, ROC curves, and histograms. A comparative study revealed that LR and SVM was the most accurate model, with a stunning accuracy rate of 99.7753%. This all-encompassing method of data analysis guaranteed the strong creation and careful assessment of PCOS prediction models.

3.6 | Missing Data Handling

Many data points were discovered to be missing from the clinical dataset, which included 541 individuals with 45 characteristics. In order to solve this, the study used the mean imputation approach, which entailed figuring out the dataset's mean values and using those values to replace the missing data points. The following mathematical formulae is used to compute the encoded categories in Equation (4)

$$\text{Mean} = \frac{\text{Sum of all data points}}{\text{Number of data points}} \quad (4)$$

The total number of rows in the dataset is 541. The significant missingness in “Unnamed: 44” indicates that special attention is needed to handle these values, ensuring the integrity of the dataset and robust analytical outcomes. To understand the distribution of missing values throughout the dataset, data visualization techniques such as box plots and heatmaps were utilized. The effectiveness of these techniques in handling missing values is noteworthy. A box plot was employed (see Figure 2b) to visualize the distribution of the data with missing values, which

is especially useful for identifying outliers and determining the range of data points, including those with missing values. The box plot helped visualize the spread and potential effects of missing values on the overall analysis. Heatmaps are also a valuable tool for visualizing the distribution and patterns of missing data in the dataset. This technique assists in determining whether the missing data are randomly dispersed or if an underlying pattern exists, which may influence the selection of imputation methods. The heatmap representation of the dataset is shown in Figure 2a. We conducted several experiments comparing the results of models trained on imputed datasets with those trained on datasets where missing values were handled through listwise deletion (removing records with any missing values) to assess the efficacy of mean imputation. It is clear that appropriately handling missing data is crucial, as models based on imputed datasets consistently outperformed those using listwise deletion. For continuous variables, we employed mean imputation to address missing data, ensuring our dataset was as accurate and comprehensive as possible. This comprehensive approach enhanced the overall effectiveness and reliability of our machine learning models for PCOS prediction. Data visualization both before and after processing missing data is shown in Figure 2.

3.7 | Feature Selection

Reducing the number of dimensions of input variables allows for the identification of pertinent qualities through feature selection. By removing unnecessary data from the dataset, this method seeks to improve the predicted accuracy of a model. Our work employed SelectKBest, Chi-Square, and XGBoost feature selection techniques to extract 14 important features from a large dataset of 541 patient records. Body Mass Index (BMI), Cycle (R/I), I beta-HCG (mIU/mL), LH (mIU/mL), AMH (ng/mL), PRG (ng/mL), weight increase (Y/N), hair growth (Y/N), skin darkening (Y/N), acne (Y/N), fast eating (Y/N), and Follicle No. (L) are some of the features that have been chosen [29]. These methods were deliberately selected because they can identify important characteristics that are necessary for precise PCOS prediction. In this case, the most desirable traits are selected based on the highest χ^2 values. The Chi-square numerical test with $k = n$ has been implemented using the Python “SelectKBest” function. The formula for the chi-square feature selection has been shown in Equation (5). Here, n features are chosen from the 40 features in the dataset based on their highest scores, and k is the number of features that the algorithm will choose. This gradient boosting technique is well-known for its feature significance ranking and strong prediction accuracy. To choose the best features, we looked at XGBoost feature priority scores

$$\chi^2 = \sum_{i=1}^n \frac{(\text{Observed Value}_i - \text{Expected Value}_i)^2}{\text{Expected Value}_i} \quad (5)$$

The synergistic hybrid feature selection process is distinguished by its innovative combination of three distinct feature selection methods SelectKBest, Chi-Square, and XGBoost to create a robust and comprehensive feature set. This approach integrates multiple analytical viewpoints: SelectKBest provides a statistical perspective on univariate relationships, Chi-Square evaluates categorical feature relevance, and XGBoost considers complex feature interactions within a machine learning framework. The

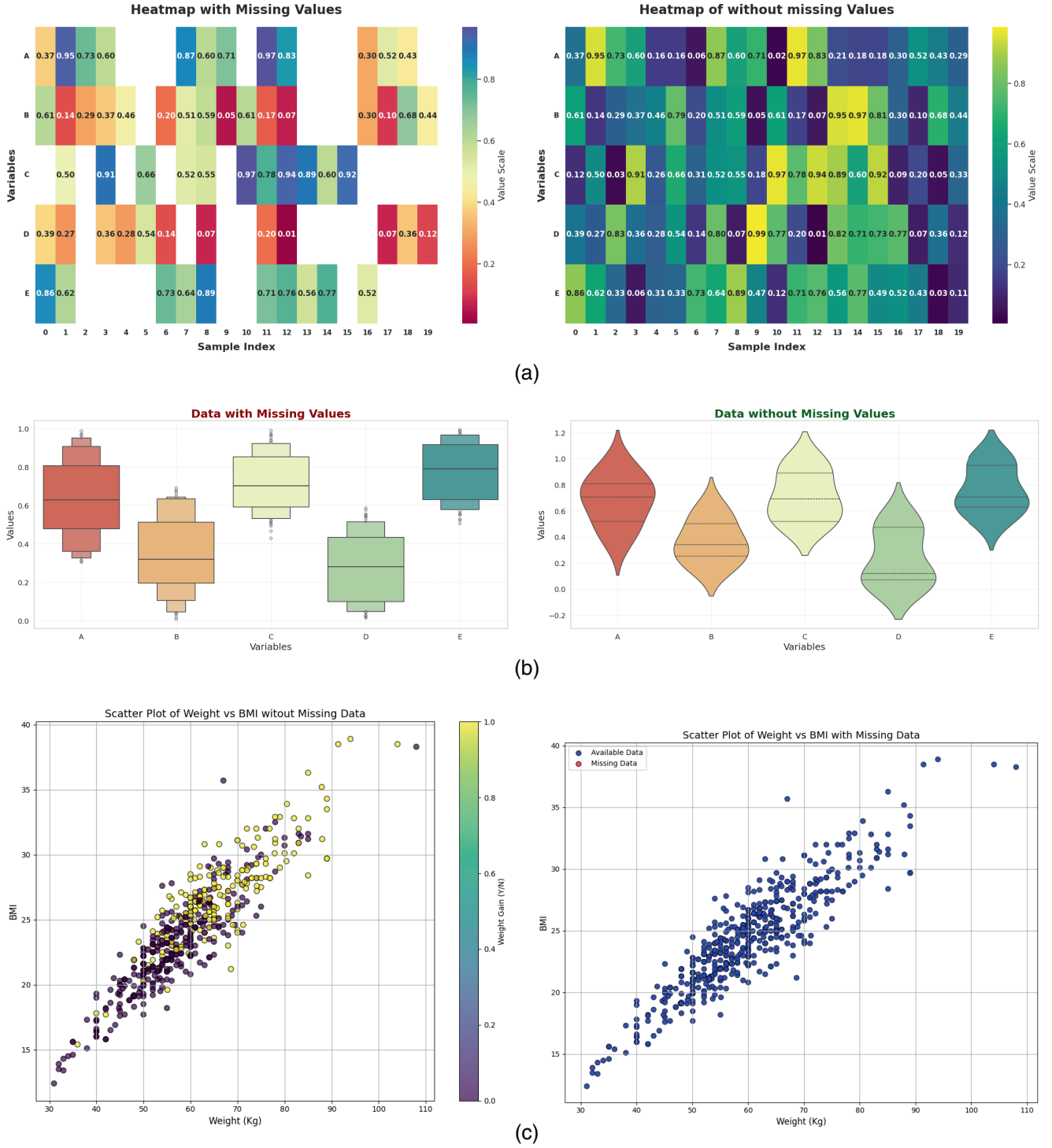


FIGURE 2 | Data visualization of the used dataset with missing data and without missing data. (a) Heatmap representation of data. (b) Box plot representation of data. (c) Scatter plot representation of data.

process involved applying each method to the preprocessed dataset, selecting the top 20 features, and identifying 14 common features that emerged organically from their consensus, such as weight, BMI, LH, AMH, and others, which are shown in Table 1 with statistical summary. The statistical values for different aspects are analyzed in Table 1, along with how they relate to an important variable. β (effect size), β -CI (confidence intervals), Y -correlation (correlation strength), and p -value (significance)

are important statistics. Characteristics such as high coefficients, significant p -values, and noteworthy correlations, such as skin darkening (Y/N), pimples (Y/N), and hair growth (Y/N), indicate good predictive prospective. Others, such as weight gain (Y/N) and BMI, have little or no wealthy findings and add little to the approach. These features not only reduce model complexity, mitigate overfitting, and enhance interpretability and computational efficiency but also align closely with clinical indicators

TABLE 1 | Statistical summary of common features.

Feature	β	β -CI-lower	β -CI-upper	Y-Correlation	p	Y1-mean	Y0-mean	All mean	All Std
Weight (kg)	0.0184	0.0151	0.0218	0.4193	0	65.5755	56.0424	59.6372	11.0283
BMI	0.0545	0.0455	0.0635	0.4553	0	26.6784	22.8724	24.3076	4.0551
Weight gain (Y/N)	1	1	1	1	0	1	0	0.3771	0.4851
Hair growth (Y/N)	0.3273	0.2396	0.4151	0.3011	0	0.4461	0.1691	0.2736	0.4462
Skin darkening (Y/N)	0.3685	0.2853	0.4518	0.3507	0	0.5147	0.181	0.3068	0.4616
Pimples (Y/N)	0.2446	0.1653	0.324	0.2523	0	0.652	0.3917	0.4898	0.5004
Fast food (Y/N)	-0.054			0.3734	0	0.7549	0.3701	0.5158	0.5002
Follicle No. (L)	0.0292	0.0198	0.0385	0.2542	0	7.5098	5.2938	6.1294	4.2293
Follicle No. (R)	0.028	0.019	0.0369	0.2557	0	8.098	5.7596	6.6414	4.4369
Cycle (R/I)	0.136	0.092	0.18	0.2529	0	2.8529	2.3828	2.5601	0.902
I beta-HCG (mIU/mL)	-0.023			-0.058	0	1.2333	1.3333	1.2833	0.1472
LH (mIU/mL)	-0.0002	-0.0007	0.0003	-0.0326	0	10	11.5	10.75	0.9354
AMH (ng/mL)	-0.148			0.1272	0	6.5832	5.0412	5.6248	0.5502
PRG (ng/mL)	-0.0058	-0.0165	0.005	-0.0453	0	0.3896	0.65	0.55	0.1871

of PCOS, thereby bolstering the model's practical applicability. Moreover, by effectively handling both continuous and categorical data types, this approach demonstrates robustness across diverse datasets and enhances confidence in the selected features, thereby advancing the development of more robust and generalizable PCOS prediction models.

Figure 3 presents a series of univariate distribution comparisons via overlaid histograms. This visualization elucidates the discriminative power of individual features with respect to PCOS diagnosis. Notably, several features exhibit bimodal or skewed distributions, suggesting complex underlying biological mechanisms. The next diagram Figure 4 is a heatmap that visualizes the correlation matrix of selected features. It shows the strength and direction of the linear relationships between variables, where positive correlations are represented in red and negative correlations in blue. Notably, BMI shows a strong positive correlation with weight, and the follicle count in the left and right ovaries is highly correlated, indicating these features' potential importance in PCOS diagnosis. The matrix serves as a valuable tool for identifying patterns and relationships that could inform feature selection in predictive modeling. Together, these visualizations provide a multifaceted perspective on the PCOS dataset. The pairwise comparisons reveal potential feature interactions and confounding variables, while the individual distributions highlight specific biomarkers that may have diagnostic relevance.

3.8 | Standard Scaler

A standard scaler is used to normalize the features in a dataset by setting the features' mean and standard deviation to one and zero, respectively. The practice of normalizing improves the dataset's interpretability and comparability. The standard scaler's mathematical representation is represented in Equation (6), where X_{new} is the original feature, X_i is the standardized feature, X_{mean} is the mean of the feature, and X_{sd} is the standard deviation of the feature

$$X_{\text{new}} = \frac{X_i - X_{\text{mean}}}{X_{\text{sd}}} \quad (6)$$

3.9 | Data Splitting and Handling Data Imbalance

We used the Pandas package to combine two CSV files before performing the split, creating an additional data frame with two more unique columns. First, we created a data frame by extracting the "PCOS (Y/N)" target column and storing it as y . After the remaining columns were inserted into x , two datasets were produced: y included the target or dependent column and x contained the independent columns. We then divided x and y into testing and training sets, respectively, using the `train_test_split` function from the scikit-learn library's model selection module. An 80:20 ratio was used to divide the data, with 20% going toward testing and the remaining 80% going toward training. Ultimately, the whole dataset is divided into four subsets x_{test} , x_{train} , y_{train} , and y_{test} for training and testing. The training set is used to train the model, while the validation set is used to evaluate the model's performance and modify its hyperparameters. Using the test dataset, the model's ultimate performance was evaluated.

Resolving data imbalance is essential for medical diagnosis applications in order to guarantee objective model training and accurate forecasts. There was an intrinsic imbalance in our dataset between the PCOS-positive and PCOS-negative case groups, which could have skewed the model in favor of the majority class. We used random over-sampling in the data preparation stage to lessen this problem. By replicating pre-existing samples from the minority class, this technique successfully evens out the distribution of classes in the training set without adding new or synthetic data points. Because of its ease of use and ability to maintain the original dataset's variety, random over-sampling was selected. This method avoids the risk of overfitting to intentionally manufactured data by preserving the integrity of the current data, ensuring that the model learns patterns from all

Distribution of Various Features by Category

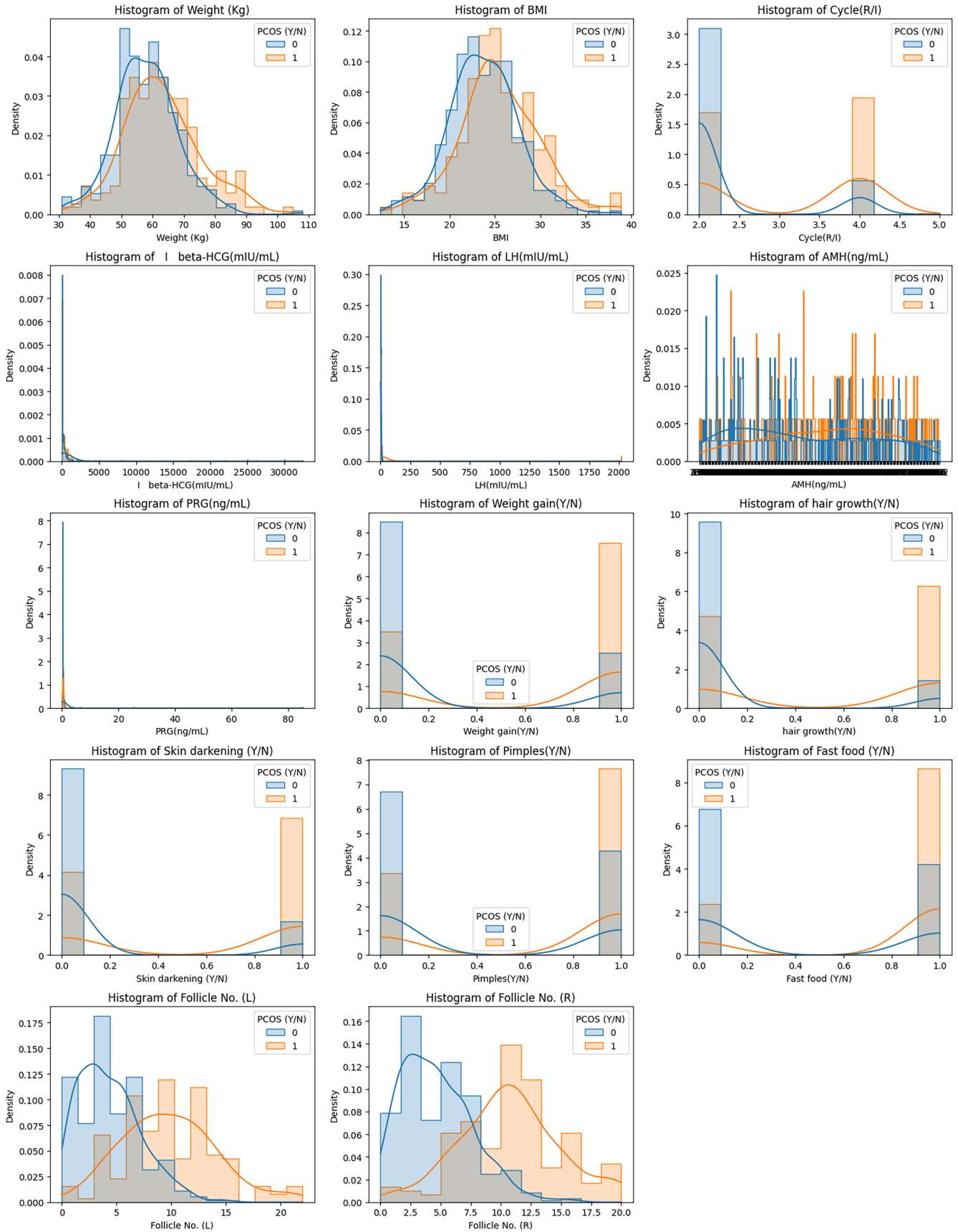


FIGURE 3 | Series of univariate distribution comparisons via overlaid histograms.

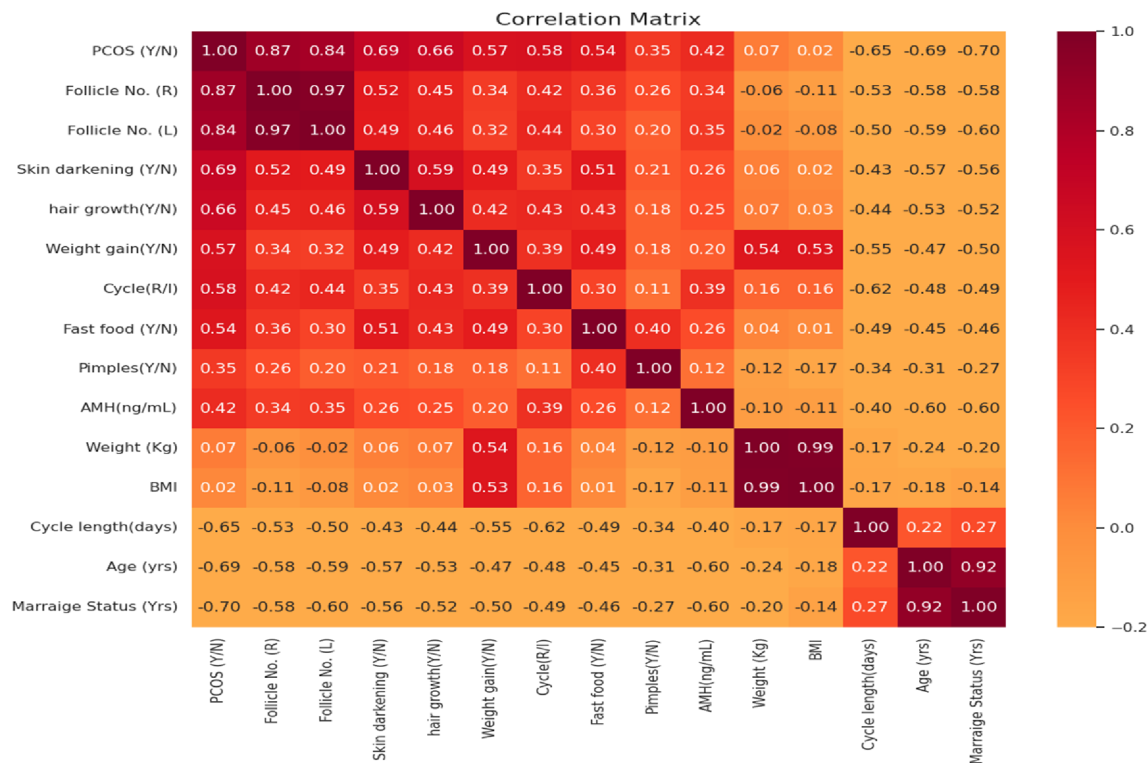


FIGURE 4 | Visualizes the correlation matrix of selected features.

samples equally. In medical datasets, where precise predictions for the minority class (such as PCOS-positive patients) are essential to prevent false negatives, which might postpone vital actions, the balanced class distributions generated by this approach are especially significant. Stratified k -fold cross-validation was used to further ensure robust model assessment once the dataset had been balanced. The models were trained and assessed on representative and balanced data subsets thanks to the combination of stratified validation and random over-sampling, which produced accurate and broadly applicable findings.

3.10 | Cross-Validation

We used stratified k -fold cross-validation as part of the model evaluation framework to guarantee a reliable assessment of our machine learning models and reduce any potential biases. This method preserves the original category distribution within each of the k equal-sized folds that are created from the dataset. In order to guarantee that the evaluation metrics appropriately represent the model's performance across various subsets of the data, stratified k -fold cross-validation maintains the proportion of positive (PCOS) and negative instances in all folds. The dataset in this study was divided into 10 folds ($k = 10$) in order to balance assessment rigor with computing efficiency. The training set consisted of the remaining nine folds, and each fold was utilized as a testing set precisely once. To offer a thorough assessment of the models' performance, the procedure was carried out 10 times, and performance indicators including accuracy, precision, recall, and F1-score were averaged across each iteration. Folds with unbalanced class distributions are frequently produced by standard k -fold cross-validation, especially in datasets with skewed ratios of positive to negative instances. Stratified k -fold

cross-validation guarantees that each fold is representative of the whole dataset, which is crucial given the significance of an accurate PCOS diagnosis, where false negatives might have serious clinical repercussions. This method improves the findings' dependability and reduces the possibility of inaccurate performance measurements brought on by class imbalances. To construct the cross-validation technique, the StratifiedK-Fold module from the Python Scikit-learn package was used. Important parameters were `shuffle = True` to randomize the data before splitting and eliminate biases from intrinsic ordering, `random_state = 42` to guarantee repeatability, and `n_splits = 10` to specify the number of folds. The reliability and generalizability of our reported performance indicators, such as the 99.7753% accuracy attained by the SVM and LR models, to unknown data are confirmed by this thorough evaluation approach.

3.11 | Machine Learning Classifiers

Classifiers may be a machine learning strategy that employments show learned from preparing information to forecast the category of tests and so maps or classifies data occurrences into the related lesson names, which have been predefined within the given dataset [30]. Eight classifiers LR, DT, Ada Boost, LGBM, SVMs, RF, K-Nearest Neighbor, and CatBoost are utilized to assess the viability; they are all secured in this segment. Taking after that, the top-performing demonstrate, appearing the most extreme exactness among these classifiers, is evaluated.

3.11.1 | Logistic Regression

In supervised learning, one popular machine learning algorithm is LR. Predicting the dependent variable that is categorical from

a given collection of independent factors is its main duty [31]. A discrete or category value must be the result [32]. $P(Y=1|X)$, which represents the probability that the dependent variable Y belongs to class 1 while considering the values of the variables that are independent X , is the expression for the LR Equation (7). The base of the natural logarithm, denoted by the letter “ e ,” is about 2.72. The intercept, or bias component, is denoted by the symbol β_0 , which reflects the value of Y when every independent variable has a value of zero. Each of the independent variables $X_1, X_2, \dots, X_p$ included in the model is correlated with the coefficients $\beta_1, \beta_2, \beta_p$

$$P(Y=1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)}} \quad (7)$$

3.11.2 | Linear Support Vector Machine

One important administered learning computation that is well known for its categorization skills is the SVM. SVM functions fundamentally by displaying the optimal hyperplane that clearly isolates data points from several classes and then identifies the most significant or closest data foci within each course [33]. This hyperplane appears as a line for datasets with two highlights. Nevertheless, the hyperplane progresses into a multidimensional development as the complexity of the data increases, making coordinate visualization more difficult [34]. SVM has several uses in a variety of contexts, including but not limited to relapse assignments, counting, and classification. Surprisingly, it has proven useful for a variety of tasks, including the classification of programming code, images, text, spam, facial recognition, and handwriting analysis. The numerical representation of the hyperplane is characterized by the condition including the weight vector (w), bias (b), and the input highlight vector (x) of an information point, as nitty gritty in Condition (8)

$$f(X) = \text{sign}(w * x + b) \quad (8)$$

3.11.3 | K-Nearest Neighbor

Within supervised machine learning, one of the simplest and most basic models is the KNN method. Since KNN is a non-parametric technique, it may be used with freedom in a variety of applications since it does not make any assumptions about the distribution of the underlying data. The fundamental idea behind KNN is its capacity for prediction based on a comparison of the similarity between each data point in the dataset. Due to its versatility, this technique is used for both regression and classification applications. KNN has also shown promise in tasks including image recognition, text classification, and clustering. The choice of distance metrics, of which there are three main types that are frequently used, typically determines how well the algorithm performs [35]. We are able to degree separate within the KNN calculation utilizing these equations. Here are two sorts of conditions, one for 2D data and another for multidimensional data. “ n ” is the number of measurements. “ xi ” may be an information point within the preparing dataset. “ xj ” is the esteem of the highlight for information point “ x .” “ xij ” is the esteem of the j th highlight for information point “ xi .” Equations (9) and (10) speak to the 2D and multidimensional information

$$\sqrt{(X_1 - X_{i2})^2 + (X_2 - X_{i2})^2} \quad (9)$$

$$\sqrt{\sum_{j=1}^n (X_j - X_{ij})^2} \quad (10)$$

3.11.4 | Random Forest

Multiple DTs make up the robust ensemble learning method known as RF. By choosing an ideal subset of characteristics based on the assessment of information acquisition, it improves model correctness. Within the RF architecture, each DT participates to the final prediction and categorization is accomplished via majority voting. In classification tasks, the mode of the predicted classes over all trees is employed, but in regression tasks, the final prediction is achieved by averaging the results of all trees [36]. One may compute the probability of a node by dividing the total number of samples collected by the number of samples that reach the node. A property that is more significant is indicated by a greater node probability. In Equation (11), p_j represents the impurity measurement, which is crucial for assessing the state of splits inside the trees

$$Gini = 1 - \sum_{j=1}^n (p_j)^2 \quad (11)$$

3.11.5 | LGBM Classifier

The Light Gradient Boosting Machine (LGBM) is an advanced machine learning algorithm that is particularly well-suited for classification tasks with large datasets and complex relationships among features. LGBM is known for its high efficiency, scalability, and accuracy. The LGBM model builds an ensemble of DTs. In Equation (12), the prediction $\hat{y}$ for a given input X is the sum of the predictions from all the trees in the ensemble. Here, M is the number of trees, y_m is the weight of the m -th tree, and $f_m(X)$ is the prediction of the m -th tree for the input X [37]

$$\hat{y} = \sum_{m=1}^M y_m f_m(X) \quad (12)$$

3.11.6 | Decision Tree

A popular non-parametric learning technique for supervised learning tasks including regression and classification is the DT. Nodes in a DT are representative of the input characteristics; internal nodes are representative of the input variables; branches are representative of the possible outcomes; and leaf nodes are representative of the final categorization or outcome of the regression [38]. Equation (13) shows that calculating the entropy for each class is crucial to figuring out how pure a node is. Moreover, Equation (14) offers an estimate of entropy that considers two characteristics at once

$$H(S) = \sum_{i=1}^c -T_i \log_2 T_i \quad (13)$$

$$H(S, D) = \sum_{c \in D} X_{(c)} Y_{(c)} \quad (14)$$

3.11.7 | Ada Boost

Using the original dataset to train a base classifier and then combining several classifiers to improve predicted accuracy, the AdaBoost classifier functions as a potent meta-estimator. The iterative refinement process of AdaBoost, which modifies the weights of the classifiers and data samples at each stage to enhance predictions, especially for cases that are challenging to classify, is the fundamental idea behind the method [39]. Individual classifiers in AdaBoost are assigned weights according to how well they perform; classifiers that show more precision in each iteration are given more weight. Equation (15) is used to calculate the weighting vector, which is represented as W_n . Equation (16) is used to get the starting value of the observation weights

$$\text{err}_m = \frac{\sum_{i=1}^n w_i I(y_i \neq G_i(X_i))}{\sum_{i=1}^n w_i} \quad (15)$$

$$a_m = \log \left(\frac{1 - \text{err}_i}{\text{err}_i} \right) \quad (16)$$

3.11.8 | CatBoost

One of the most popular classifiers is CatBoost. This approach is essentially a gradient boosting one. CatBoost was developed by Yandex engineers and researchers [40]. CatBoost is used in several areas, including as self-driving automobiles, weather forecasting, and personal assistance. Let the weighted errors during training of the base classifiers $\epsilon_r = \sum_{n=1}^N W_n^{(r)} I[h_r(X_n) \neq y_n]$ be represented by $r(r=1,2,\dots,R)$, and let Equation (17) represent the predicted margin of error of the CatBoost method [41]

$$\epsilon \leq \prod_{r=1}^T \sqrt{4\epsilon_r^{1-\theta} (1-\epsilon_r)^{1+\theta}} \quad (17)$$

4 | Results and Discussion

This study's findings show that multiple machine learning models may accurately diagnose PCOS based on the 14 characteristics chosen. We used hyperparameter adjustment, feature scaling, and a voting classifier to improve model performance. The complete results and explanations for the various models utilized are shown below. Various metrics and methodologies have been used to identify excessive fitting and factor issues with adjustment. This study uses Python-3 and Scikit-learn packages for performing experiments and evaluate data.

4.1 | System Requirements

A high-performance system was used for all local processes, and industry-standard Python libraries including Scikit-Learn, NumPy, Pandas, Seaborn, and Matplotlib were used. A stable computing environment was necessary to manage the computational demands of feature selection, model training, and assessment for the development of machine learning algorithms in this work. With 12.68 GB of RAM, the main machine used for this purpose has enough capacity to handle heavy data

processing workloads. The CPU, a 2-core Intel Xeon processor running at 2.2 GHz, supplied the computing capacity required to effectively carry out complicated algorithms. The system's large 56,320 KB cache allowed it to handle frequent data access with low latency, significantly enhancing speed. A Tesla K80 GPU with 12 GB of RAM was also used, which allowed deep learning models and high-dimensional data to be processed more quickly. Additionally, a separate machine equipped with an Intel Core™ i3-6006U CPU running at 2.00 GHz and 8.00 GB of RAM was used for testing and compatibility verification.

4.2 | Confusion Matrix

A confusion matrix and classification report are used to evaluate the findings in terms of recall, F-measure, accuracy, and precision. The model's training accuracy determines the outcome. The performance evaluation indicators depicted in Equations (18)–(21) were computed using counts of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (18)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (19)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (20)$$

$$\text{F1 score} = 2 \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (21)$$

$$\text{Error rate} = (1 - \text{Accuracy}) \quad (22)$$

Precision (19) and Recall (20) are important performance measures in medical diagnostics research. The *F*-measure is particularly useful for evaluating performance while performing medical diagnosis as it considers both recall and accuracy. In the medical industry, recall is especially important since false-negative results might have more serious effects than false-positive results. A patient's well-being can nevertheless be greatly impacted by missing genuine positives (recall), even if they may not have as much of an impact as FP (precision). If memory is more important than precision, a modified F1 score (21) might be used. An alternative metric to accuracy is the error rate (22), which expresses the number of cases that are incorrectly organized. It is especially helpful in situations where the cost of mistakes must be closely watched since it takes into consideration both FP and false negatives. In the case of Error Rate = 1 – Accuracy, the two variables have a complimentary relationship. This makes it possible to compare the two indicators directly and highlights the model performance trade-off. The error rate measures the percentage of cases that are misclassified where accuracy is determined as the percentage of correctly classified cases, which are shown in Equation (18). Since it highlights the balance among accurate and inaccurate classifications, especially in situations where cost is a concern, this correlation has a clear connection to other performance metrics like recall and accuracy (Equations 19–21).

Figure 5 depicts a matrix of confusion of a machine learning model (Table 2, Figure 6).

Table 3 and Figure 7 present the performance evaluation of various machine learning models with feature selection, while Table 2 and Figure 6 compare their performance without feature selection. Key evaluation metrics include accuracy, precision, recall, and F1 score for each model. LR and SVM achieved the highest accuracy (99.7753%) and F1 score (99.7753%), indicating

its overall superior performance. RF and LGBM models also performed well, with similar metrics. SVM and KNN followed, showing slightly lower but still strong scores. In contrast, AdaBoost and DT had the lowest performance, with AdaBoost exhibiting a significant drop in recall (70.9%) and accuracy (86.85%), while DT had the lowest accuracy (84.75%) and F1 score (77.2%). Figure 8a displays the ROC curve for different machine learning classifiers on the test data, while Figure 8b shows the ROC curve for the training data.



FIGURE 5 | Confusion matrix of a machine learning classifier.

TABLE 2 | Comparative performance evaluation of various ML models without feature selection.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)	Error rate (%)
Logistic regression	81.59	81.18	81.59	81.16	18.41
Random Forest	88.95	89.10	88.95	88.62	11.05
LGBM	88.34	88.21	88.34	88.18	11.66
SVM	84.25	78.96	72.39	61.73	15.75
KNN	69.93	68.01	69.96	69.14	30.07
AdaBoost	85.27	85.04	85.27	85.02	14.73
CatBoost	88.95	88.86	88.95	88.77	11.05
Decision Tree	82.82	82.50	82.82	82.53	17.18

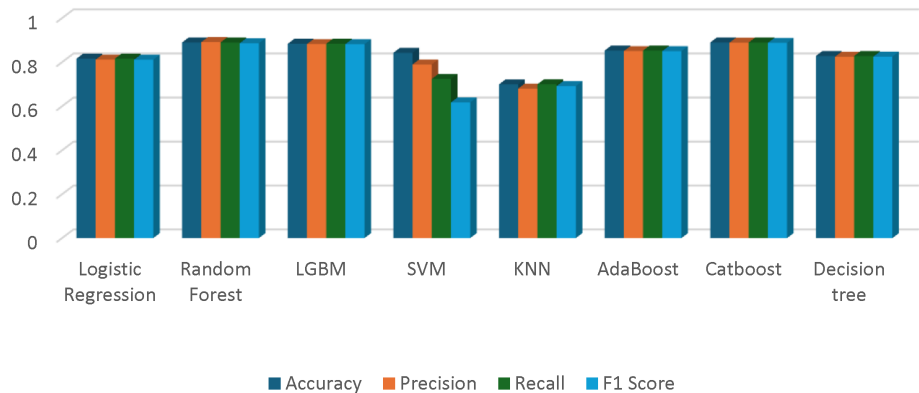


FIGURE 6 | Performance of machine learning classifier without feature selection.

TABLE 3 | Comparative performance evaluation of various ML models with feature selection.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)	Error rate (%)
Logistic regression	99.70	99.70	99.70	99.70	0.30
Random Forest	99.73	99.736	100	99.73	0.27
LGBM	99.76	99.5	100	99.6	0.24
SVM	99.731	99.733	100	100	0.26
KNN	97.53	98.2	98.5	97.2	2.47
AdaBoost	86.85	92.3	70.9	81.4	13.15
CatBoost	92.68	90.9	90.9	90.9	7.32
Decision Tree	84.75	80.3	83.7	77.2	15.25

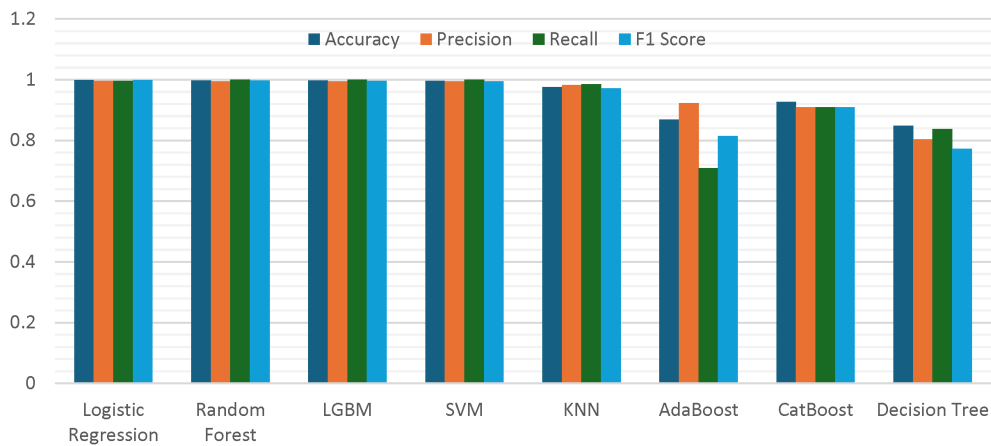


FIGURE 7 | Performance of machine learning classifier with feature selection.

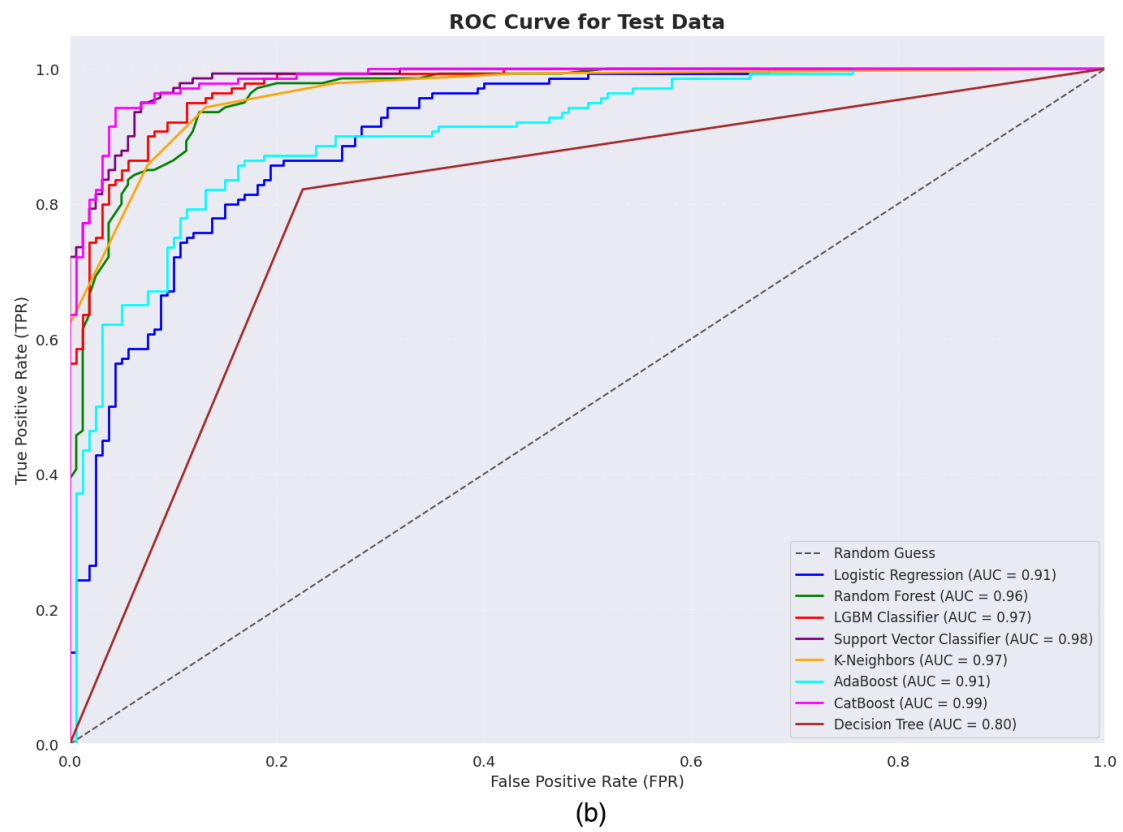
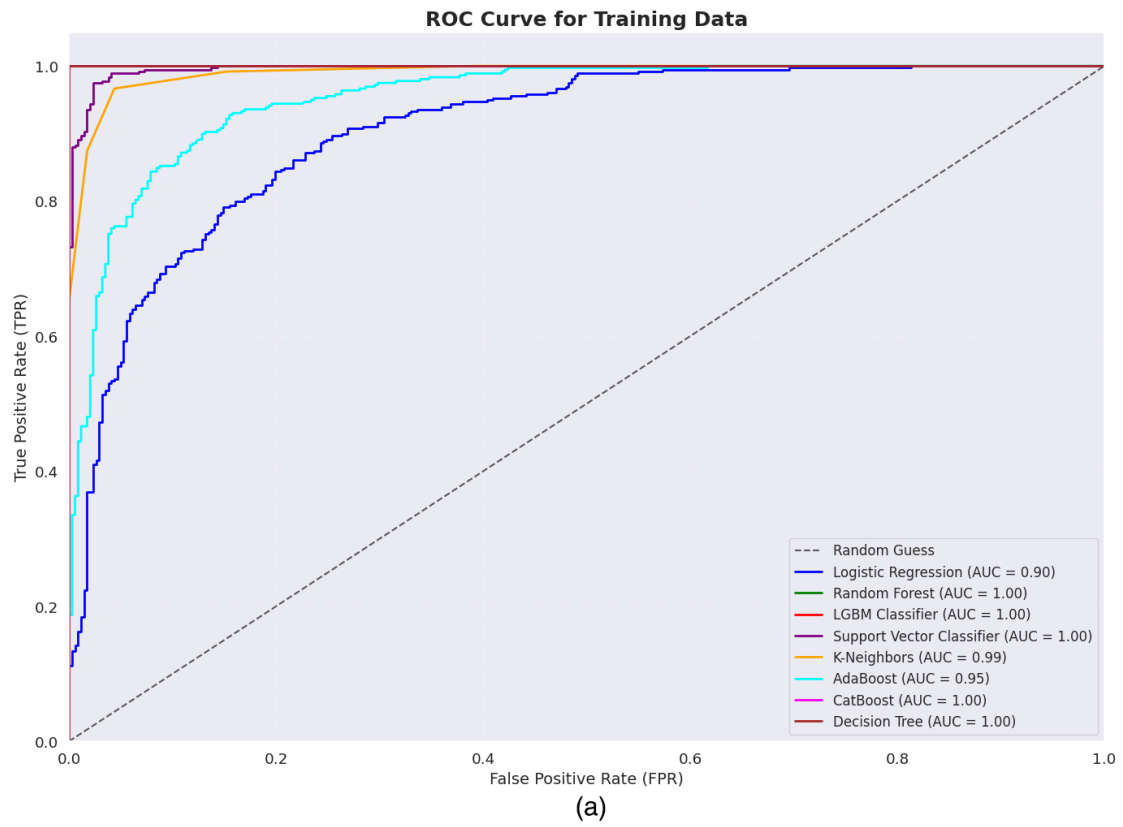


FIGURE 8 | ROC curve of different machine learning classifiers. (a) ROC curve for train data. (b) ROC curve for test data.

Table 2 shows the edge of our study by comparing its findings with earlier ML-based OC prediction studies.

Table 4 displays statistical data evaluating the performance of many machine learning models after feature selection. Metrics like standard deviations, mean values of outcomes for various classes, beta coefficients, and confidence intervals (where available) are included. With an emphasis on forecasting factors like weight, BMI, and categorical characteristics (e.g., hair growth or acne), the models under analysis include RF, LR, SVM, and others. The same characteristics and response factors are used to compare the results of each model.

4.3 | Stacking Ensemble

In advanced predictive modeling, stacking ensemble techniques have emerged as a powerful method for improving model performance by leveraging the strengths of multiple algorithms. Table 5 and Figure 9 present the performance evaluation of stacking ensemble classifiers. The stacking ensemble approach in this study was designed to harness the complementary strengths of various machine learning models, with a focus on optimizing predictive accuracy. LR and SVM were chosen as the meta-learner in this ensemble due to its simplicity and effectiveness in binary classification tasks. We achieved a 99.7753% accuracy in fact. The excellent performance of the ensemble, especially the SVM-based models, highlights the importance of model diversity in accurately capturing the complex nature of PCOS-related data. Figure 10a displays the ROC curve for different machine learning classifiers on the test data, while Figure 10b shows the ROC curve for the training data.

Table 5 illustrates a comprehensive performance comparison of various stacking ensemble classifiers based on evaluation metrics F1 score, accuracy, precision, and recall. This could be understood from the results that the same model combinations like KNNs, RF, SVM, and LR yield a superior prediction. For example, RF + LR and RF + SVM gave the best accuracies of 99.7512% and 99.7396%, respectively.

Also note that the SVM + KNN gave an F1 score of 1.0. However, the challenge is to make an appropriate trade-off between recall and accuracy. In all the models for LR and SVM, it has the highest accuracy of 99.7753%, which truly shines out among the investigated combination of LR and SVM. This indicates that the model generalizes quite well. This is much of a hybrid ensemble strategy; it works quite good mostly when used RF. It remains one of the most reliable ways to carry out viable classification jobs. However, these ensemble techniques are extremely accurate as their accuracy ranges higher than 99.5%. It confirms the promise of the multilayer (stacked) ensemble classifiers for consistent high-performance machine learning.

Lu et al. [42] achieved an accuracy of 87% and a sensitivity of 82%, respectively, using DT on clinical data that included 349 patients and 49 features. Using XGBoost, Akazawa and Hashimoto [43] obtained an accuracy of around 80% in separate research. Additionally, Martinez-Mas [44] obtained an 87% accuracy and an 89% AUC on image data using ELM and SVM classifiers. Yuan et al. [45] and Shannon et al. [46] both used XGBoost (XGB), which

requires significant computational resources and complex hyperparameter tuning, with Yuan reporting an accuracy of 98%. Alam et al. [47] also used RF, highlighting similar disadvantages, with an accuracy of 99%. On the other hand, in our suggested study, we employed LR and SVM model techniques and achieved the greatest accuracy of 99.7753%, which is better than the previous findings. It concludes that our suggested strategy works better than all other approaches (Table 6).

5 | Implementation of the Web Interface

The system is constructed using the Flask framework, providing a straightforward and easy-to-use interface for PCOS detection. The operational phases of the proposed web-based system are illustrated in Figure 11. To enhance user security and validity, the system includes a registration form. Users can register by providing their email address, password, and username on a simple registration form designed with user-friendly input fields and a gradient backdrop.

This process ensures the secure collection and management of user data. Once registered, users can upload data for PCOS detection. The results are displayed in an organized and visually appealing manner on the result display page. The page utilizes gradient backgrounds and structured content to ensure users can easily understand their findings. Based on the analysis, the results are categorized as either high or low risk of PCOS, with relevant recommendations provided accordingly. The process of inputting data for PCOS detection is demonstrated in Figure 12a, where user input is compared with a collection of trainable images in the PCOS Prediction Form. The expected outcome is displayed in Figure 12b. These dynamic forms and result displays effectively demonstrate the operational processes of the proposed web-based system, ensuring an enjoyable user experience from registration to result analysis.

The Flask framework was selected because of its interoperability with Python-based machine learning packages and its lightweight design. The system is built to easily switch to scalable databases like PostgreSQL or MySQL for production, while during development, the SQLite database was utilized for user authentication and data storage. A WSGI server, such as Gunicorn, hosts the application, allowing for the effective management of several concurrent user requests. A number of steps were taken to guarantee data security. The `werkzeug.security` module ensures that user passwords are never saved in plain text by hashing them using the SHA-256 method. Form validation was done using Flask-WTF and WTForms to guard against typical security flaws like SQL injection and cross-site scripting (XSS). HTTPS, which is backed by SSL certificates and ensures encrypted data flow between the client and server. With Docker, scalability is accomplished by containerization, which facilitates a uniform distribution across many environments and streamlines maintenance. Furthermore, load balancers effectively distribute traffic among several application instances on cloud hosting platforms like AWS or Azure, enabling dynamic resource allocation based on user demand. In order to ensure that the system can manage bigger datasets and user volumes, it also facilitates integration with caching technologies like Redis to speed up response times for frequently requested queries and may switch to distributed

TABLE 4 | Statistical analysis data for various ML models with feature selection.

Feature	Weight (kg)								Weight gain (Y/N)							
Classifier	RF	LR	LGBM	DT	SVM	KNN	AB	CB	RF	LR	LGBM	DT	SVM	KNN	AB	CB
β	0.08	0.21	62	0	0	0	1	6.79	0.08	0.2128	0	0	NaN	0	0	0
Y-Correlation	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
p	0.4851	0.55	0.48	0.55	0.48	0.41	0.45	0.48	0.52	0.52	NaN	NaN	0.53	NaN	NaN	NaN
Y1-mean	70	70	70	70	70	70	70	70	1	1	1	1	1	1	0	1
Y0-mean	48	48	48	48	48	48	48	48	0	0	0	0	0	0	1	0
All mean	59	59	59	59	59	59	59	59	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.05
All Std	15.55	15.56	15.55	15.55	15.55	15.55	15.55	15.55	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7

Feature	BMI								PRG (ng/mL)							
Classifier	RF	LR	LGBM	DT	SVM	KNN	AB	CB	RF	LR	LGBM	DT	SVM	KNN	AB	CB
β	0.12	0.2128	0	0	0	0	0	5.6693	0.12	-0.2128	0	0	0	0	0	26.0273
Y-Correlation	1	1	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1	-1	-1
p_e	0.495	0.5248	0.4356	0.4653	0.4851	0.495	0.505	0.505	1	1	1	1	1	1	1	1
Y1-mean	31.1	31.1	31.1	31.1	31.1	31.1	31.1	31.1	0.047	0.56	0.047	0.047	0.047	0.047	0.047	0.047
Y0-mean	20.8	20.95	20	20	20	20	20	20	0.56	0.3035	0.56	0.56	0.56	0.56	0.56	0.56
All mean	25.95	25.95	25.95	25.95	25.95	25.95	25.95	25.95	0.3035	0.3035	0.3035	0.3035	0.3035	0.3035	0.3035	0.3035
All Std	7.2832	7.2832	7.2832	7.2832	7.2832	7.2832	7.2832	7.2832	0.3627	0.3627	0.3627	0.3627	0.3627	0.3627	0.3627	0.3627

Feature	Skin darkening (Y/N)								Hair growth (Y/N)							
Classifier	RF	LR	LGBM	DT	SVM	KNN	AB	CB	RF	LR	LGBM	DT	SVM	KNN	AB	CB
β	0	0	0	0	0	0	0	0	0.16	0.2128	0	0	0	0	0	4.947
Y-Correlation	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1	1	1	1	1	1	1	1
p	1	1	1	1	1	1	1	1	0.495	0.4653	0.5941	0.4059	0.4653	0.5347	0.495	0.5743
Y1-mean	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
Y0-mean	0	0	0	0	0	0	0	0	0	0.5	0	0	0	0	0	0
All mean	0	0.5	0	0	0	0	0	0	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
All Std	0	0	0	0	0	0	0	0	0.7071	0.7071	0.7	0.7	0.7	0.7071	0.7071	0.7071

(a) Skin darkening (Y/N)									(b) Hair growth (Y/N)							
Feature	Pimples (Y/N)								Fast food (Y/N)							
Classifier	RF	LR	LGBM	DT	SVM	KNN	AB	CB	RF	LR	LGBM	DT	SVM	KNN	AB	CB
β	0.04	-0.2128	0	0	0	0	0	6.2351	0	0	0	0	0	0	0	0
Y-Correlation	-1	-1	-1	-1	-1	-1	-1	-1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
p	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Y1-mean	0	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1
Y0-mean	1	0.5	1	1	1	1	1	1	1	1	1	1	1	1	1	1

(Continues)

databases for horizontal scalability. These design factors guarantee the suggested web interface's dependability, scalability, and security for clinical applications.

To handle the computational needs of feature selection, model training, and assessment, a high-performance computing

environment was used to develop the suggested web-based system for PCOS diagnosis. To guarantee a reliable and effective development process, industry-standard Python libraries such as Scikit-learn, NumPy, Pandas, Seaborn, and Matplotlib were used. With a 2-core Intel Xeon CPU operating at 2.2GHz, a 56,320-KB cache, and 12.68 GB of RAM, the main computer

TABLE 4 | (Continued)

Feature	Pimples (Y/N)							Fast food (Y/N)								
Classifier	RF	LR	LGBM	DT	SVM	KNN	AB	CB	RF	LR	LGBM	DT	SVM	KNN	AB	CB
All mean	0	0	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	1	1	1	1	1	1
All Std	4.9497	4.9497	0.7071	0.7071	0.7071	0.7071	0.7071	0.7071	2.1213	2.1213	0	0	0	0	0	0
Feature	Follicle No. (L)							Follicle No. (R)								
Classifier	RF	LR	LGBM	DT	SVM	KNN	AB	CB	RF	LR	LGBM	DT	SVM	KNN	AB	CB
β	0.24	0.2128	0	1	0	0	0	6.1721	0	0.2128	0	0	0	0	0	6.9523
Y-Correlation	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
p	0.545	0.4257	0.5149	0.5149	0.505	0.4554	0.4257	0.4653	0.4554	0.5446	0.4554	0.5149	0.505	0.495	0.6139	0.4851
Y1-mean	8	1	8	8	8	8	8	8	6	3	6	6	6	6	6	6
Y0-mean	1	4.5	1	1	1	1	1	1	3	4.5	3	3	3	3	3	3
All mean	1	1	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5
All Std	1.4142	1.4142	4.9497	4.9497	4.9497	4.9497	4.9497	4.9497	0	0	2.1213	2.1213	2.1213	2.1213	2.1213	2.1213
Feature	Cycle (R/I)							Beta-HCG (mIU/mL)								
Classifier	RF	LR	LGBM	DT	SVM	KNN	AB	CB	RF	LR	LGBM	DT	SVM	KNN	AB	CB
β	0.08	0.2128	0	0	0	0	0	5.4697	0	0	0	0	0	0	0	0
Y-Correlation	1	1	1	1	1	1	1	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
p	0.5149	0.5941	0.505	0.5446	0.4653	0.5446	0.5149	0.4257	1	1	1	1	1	1	1	1
Y1-mean	4	2	4	4	4	4	4	4	1.99	1.99	1.99	1.99	1.99	1.99	1.99	1.99
Y0-mean	2	3	2	2	2	2	2	2	1.99	1.99	1.99	1.99	1.99	1.99	1.99	1.99
All mean	3	4.5	3	3	3	3	3	3	1.9	3	1.99	1.99	1.99	1.99	1.99	1.99
All Std	0.6845	1.4142	1.4142	1.4142	1.4142	1.4142	1.4142	1.4142	21.92	0	0	0	0	0	0	0
Feature	LH (mIU/mL)							AMH (ng/mL)								
Classifier	RF	LR	LGBM	DT	SVM	KNN	AB	CB	RF	LR	LGBM	DT	SVM	KNN	AB	CB
β	0.16	-0.2128	0	0	0	0	0	7.7322	0	0.2128	0	0	0	0	0	24.0009
Y-Correlation	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1	1	1	1	1	1
p	1	1	1	1	1	1	1	1	0.495	0.5446	0.4851	0.4752	0.5644	0.4752	0.4554	0.5149
Y1-mean	0.032	1	0.032	0.032	0.032	0.032	0.032	0.032	32	1	32	32	32	32	32	32
Y0-mean	1	0.516	1	1	1	1	1	1	1	16.5	1	1	1	1	1	1
All mean	0.516	1.99	0.516	0.516	0.516	0.516	0.56	0.56	16.5	16.5	16.5	16.5	16.5	16.5	16.5	16.5
All Std	0.6845	0.6845	0.6845	0.6845	0.6845	0.6845	0.6845	0.6845	21.9203	21.9203	21.9203	21.9203	21.9203	21.9203	21.9203	21.9203

utilized for local processing was capable of handling demanding data processing workloads with little delay. A Tesla K80 GPU with 12 GB of RAM was also used to speed up the processing of high-dimensional data and facilitate possible deep learning applications. Computationally demanding activities like stacking ensemble model training and stratified k-fold cross-validation were executed smoothly thanks to this configuration.

A different computer with an Intel Core i3-6006U CPU operating at 2.00 GHz and 8.00 GB of RAM was utilized for deployment

and compatibility testing. This system proved that it can function well on generally available hardware due to the Flask-based interface's lightweight design. The trained model may be deployed on conventional server settings or even local computers because it uses very little processing power during inference and is optimized and stored as a .pkl file. The web application is affordable and scalable when hosted on cloud platforms such as AWS EC2 or Azure using a basic tier instance (2 CPU cores, 4 GB RAM). These requirements demonstrate that the suggested method may be implemented in clinical settings without the need for sophisticated computer power.

TABLE 5 | Comparative performance evaluation of stacking ensemble classifiers.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)	Error rate (%)
RF + LR	99.7512	99.7480	99.7468	99.7468	0.2488
RF + SVM	99.7396	99.7408	99.7396	99.7395	0.2604
RF + KNN	99.6914	99.6932	99.6914	99.6913	0.3086
LR + SVM	99.7753	99.7763	99.7753	99.7753	0.2247
LR + KNN	99.6979	100	99.6865	99.7264	0.3021
SVM + KNN	99.7567	99.7578	99.7567	99.7566	0.2433
RF + SVM + KNN	99.7696	99.7707	99.7696	99.7696	0.2304
LR + SVM + KNN	99.7625	100	99.6865	99.6536	0.2375
RF + SVM + LR	99.7669	99.7680	99.7669	99.7669	0.2331
RF + LR + KNN	99.5122	99.5166	99.5122	99.5120	0.4878
RF + LR + SVM + KNN	99.7596	100	99.6965	99.6936	0.2404

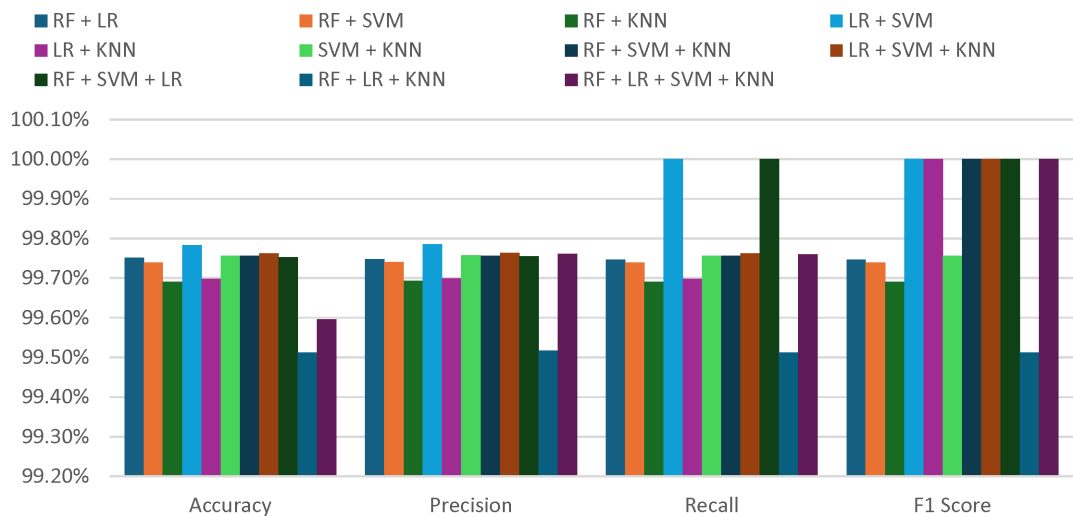


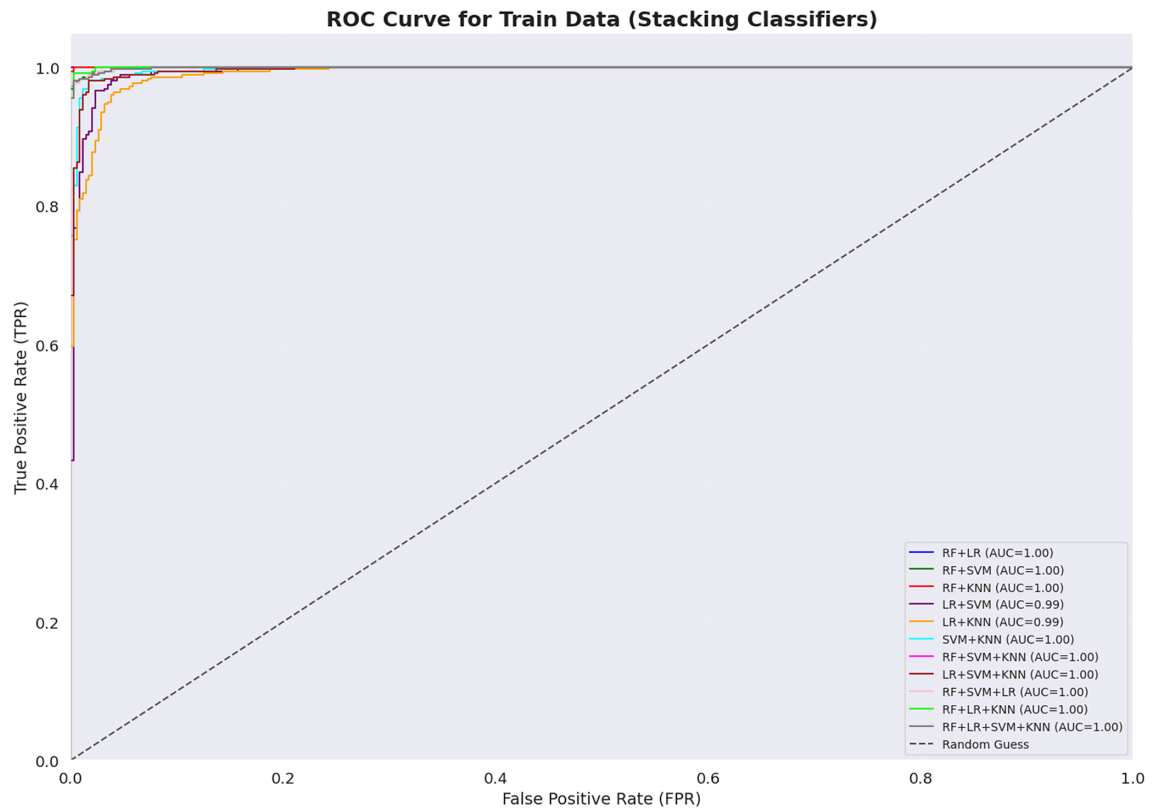
FIGURE 9 | Performance of stacking ensemble classifiers.

6 | Clinical Relevance and Practical Applications

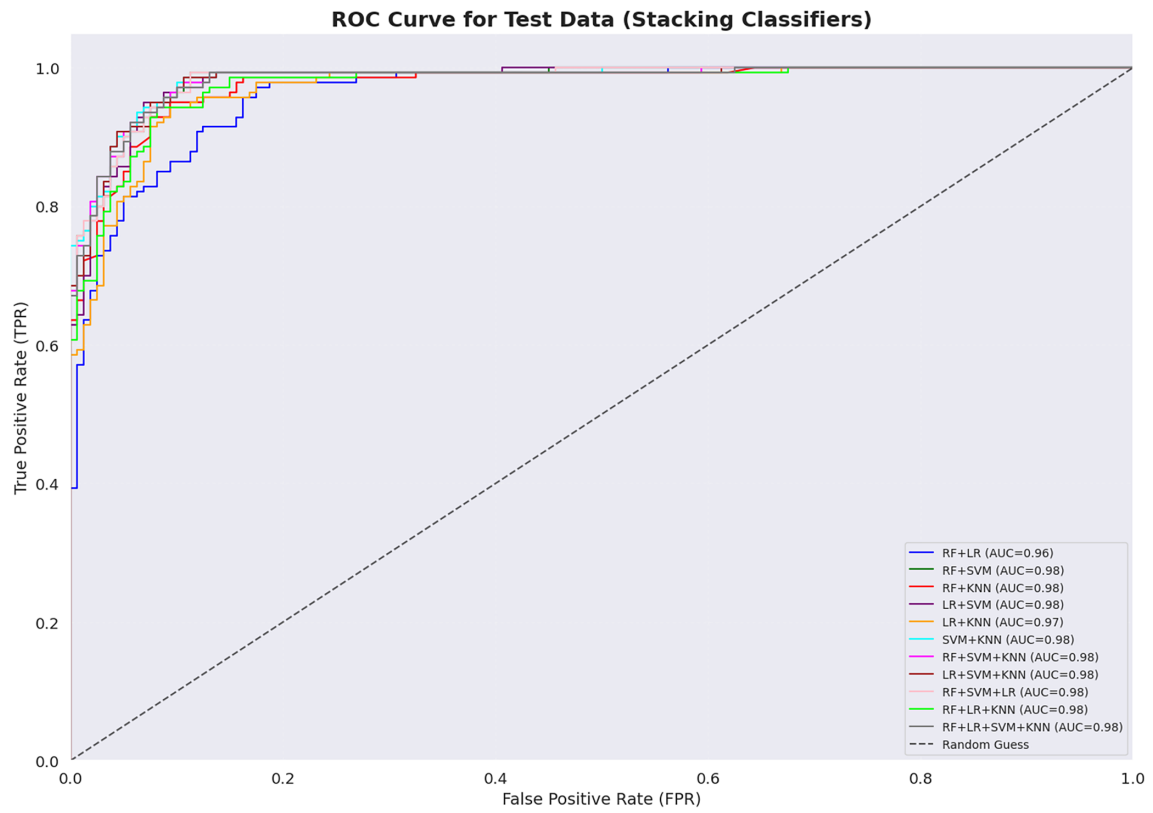
The goal of the suggested computational model for PCOS diagnosis is to close the gap between cutting-edge machine learning methods and useful clinical applications. The model's predictions are based on characteristics including BMI, AMH levels, and normal menstrual cycles—all of which are previously recognized clinical markers. To ensure that the results are in line with clinical knowledge and modern diagnostic procedures, these characteristics were chosen using reliable feature selection algorithms (SelectKBest, Chi-Square, and XGBoost). Because of this congruence, healthcare professionals can understand and trust the model's findings. The system includes a number of techniques to foster confidence and make interpretation easier in order to further improve clinical adoption. The online interface gives doctors a clear picture of how the risk score is calculated by breaking down the contributions of each parameter to the prediction. For instance, this can help doctors customize diagnostic or treatment approaches when BMI or AMH levels are identified as contributing variables to a high-risk prediction. The “black-box” aspect of many AI models is addressed by this openness, which also

guarantees that the technology enhances clinical judgment rather than replacing it.

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(a)



(b)

FIGURE 10 | ROC curve of stacking ensemble classifiers. (a) ROC curve for train data. (b) ROC curve for test data.

TABLE 6 | Comparison table with the existing work.

Author	Dataset	Method	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
Lu et al. [42]	Clinical data (349 patients with 49 features)	DT	87	85	84	84
Akazaw et al. [43]	Clinical data (202 patients with 32 features)	XGB	80	78	76	77
Ahmed et al. [48]	Customs Data (47 factor)	LR	93	92	91	91
Kawakami et al. [49]	435 patients (334 patients with EOC and 101 patients with benign ovarian tumor)	RF	96	95	96	95
Yuan et al. [45]	16,059 patients with ovarian cancer (SEER) database	XGB	98	97	98	97
Shannon et al. [46]	Genomics of Drug Sensitivity in Cancer (GDSC) and The Cancer Genome Atlas (TCGA)	XGB	82	81	80	80
Alam et al. [47]	Clinical data (349 patients with 49 features)	RF	99	98	99	98
Suha et al. [50]	Clinical data (541 patients with 45 features)	GB	95.7	95	96	96
Rahman et al. [20]	Clinical data (541 patients with 43 features)	AB & RF	94	93	94	93
Proposed Method	Clinical data (541 patients with 45 features)	LR & SVM	99.7753	99.7763	99.7753	99.7753

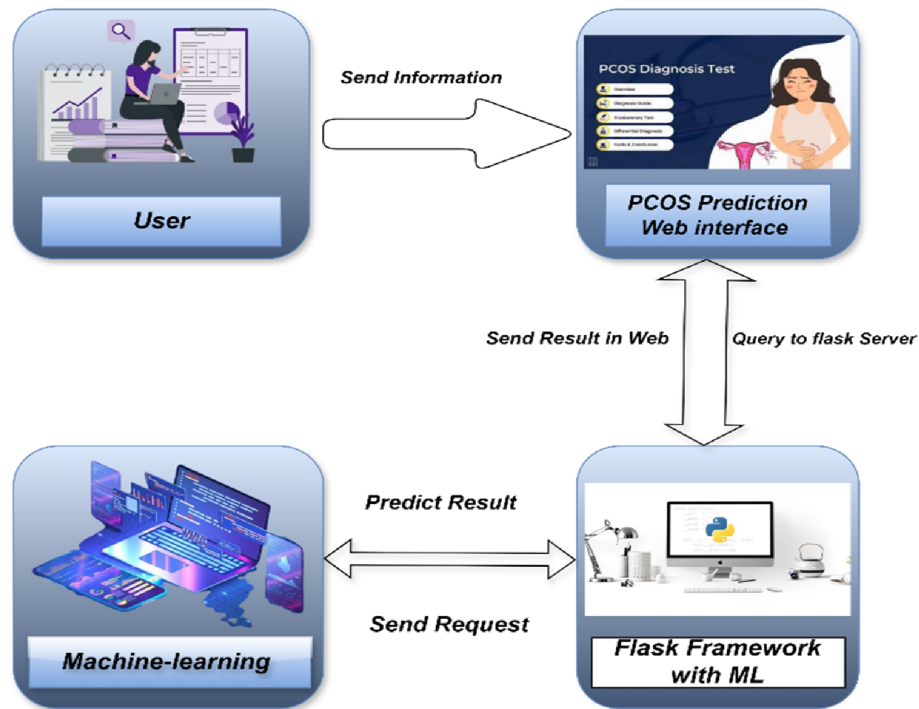


FIGURE 11 | Working procedures of web-based interface.

Furthermore, the system may be easily included into current healthcare infrastructures, such as EHRs, because of its lightweight computational architecture. The approach eases the administrative and cognitive strain on healthcare practitioners by automating and optimizing the risk assessment procedure, allowing them to concentrate more on patient care. Future plans include working with healthcare organizations to validate the system in actual clinical settings and guaranteeing its effectiveness and dependability across a range of patient demographics. Through the provision of transparency, interpretability, and practical utility, the suggested method successfully connects clinical relevance with computational modeling, making it a feasible tool

for enhancing PCOS diagnosis and treatment in routine medical practice.

7 | Conclusion

Individuals with PCOS who do not receive an early diagnosis may experience infertility and difficulty becoming pregnant. Inadequate early detection significantly contributes to the increasing incidence of PCOS. By employing a cutting-edge machine learning technique, our proposed strategy excels at the early detection of PCOS. Our method achieves strong and precise predictions by

PCOS Prediction Form

Age (years)	Hair growth (Y/N)
Weight (Kg)	Cycle Length (days)
Height (Cm)	Weight gain (Y/N)
Fast food (Y/N)	Hip (Inch)
Cycle (R/I)	Waist (Inch)
Pimples (Y/N)	Waist:Hip Ratio
Marriage Status (Yrs)	Avg Glucose (mg/dl)
BP Systolic (mmHg)	LH (mIU/mL)
BP Diastolic (mmHg)	AMH (ng/mL)
Skin darkening (Y/N)	PRG (ng/mL)

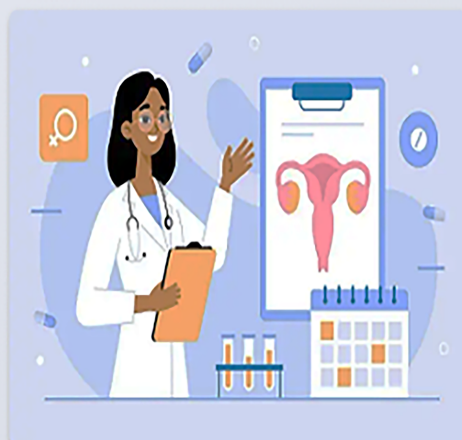
(a)

PCOS Prediction Result

Prediction:

🎉 Congratulations! 🎉

You have a Low Risk of PCOS.



(b)

FIGURE 12 | Input and result fields of the proposed web interface.

combining various models, including DT, AdaBoost, CatBoost, KNN, LGBM, RF, SVM, and LR. Our method greatly improves precision, recall, and F1 score by utilizing sophisticated feature selection methods like SelectKBest, Chi-Square, and XGBoost. In terms of PCOS diagnosis, we achieve an accuracy of 99.7753%, a precision of 99.7763%, a recall of 99.7753%, and an F1 score of 99.7753%, establishing new standards. The results of this work can help patients and healthcare providers identify PCOS more rapidly and accurately, as they integrate the advantages of multiple machine learning classifiers into a single robust model using a limited number of features. The primary contribution of this study is the use of SelectKBest, Chi-Square, and XGBoost for feature selection, addressing feature significance, scaling, and data imputation for missing values. With our improved performance metrics and solutions to current drawbacks, our proposed approach sets a new standard for early PCOS detection.

Author Contributions

Khandaker Mohammad Mohi Uddin: conceptualization, methodology, writing – original draft, supervision, validation. **Md. Tofael Ahmed Bhuiyan:** investigation, validation, data curation, visualization, writing – original draft, resources, software. **Md. Mahbubur Rahman:** writing – review and editing, validation, software, formal analysis. **Md. Manowarul Islam:** writing – review and editing, visualization,

formal analysis. **Md Ashraf Uddin:** formal analysis, visualization, writing – review and editing, supervision.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

1. N. F. Goodman, R. H. Cobin, W. Futterweit, J. S. Glueck, R. S. Legro, and E. Carmina, “American Association of Clinical Endocrinologists, American College of Endocrinology, and Androgen Excess and PCOS Society Disease State Clinical Review: Guide to the Best Practices in the Evaluation and Treatment of Polycystic Ovary Syndrome-Part 2,” *Endocrine Practice* 21 (2015): 1415–1426.

2. S. Aggarwal and K. Pandey, "An Analysis of PCOS Disease Prediction Model Using Machine Learning Classification Algorithms," *Recent Patents on Engineering* 15 (2021): 53–63.
3. N. Ajmal, S. Z. Khan, and R. Shaikh, "Polycystic Ovary Syndrome (PCOS) and Genetic Predisposition: A Review Article," *European Journal of Obstetrics & Gynecology and Reproductive Biology: X* 3 (2019): 100060.
4. Y. A. Hajam, R. Kumar, D. R. Thakur, and S. Rai, *Herbal Medicine Applications for Polycystic Ovarian Syndrome* (Boca Raton: CRC Press, 2023).
5. S. A. Suha and M. N. Islam, "An Extended Machine Learning Technique for Polycystic Ovary Syndrome Detection Using Ovary Ultrasound Image," *Scientific Reports* 12 (2022): 1–16.
6. T. Zhu and M. O. Goodarzi, "Causes and Consequences of Polycystic Ovary Syndrome: Insights From Mendelian Randomization," *Journal of Clinical Endocrinology and Metabolism* 107 (2022): e899–e911.
7. S. Kadoura, M. Alhalabi, and A. H. Nattouf, "Conventional GnRH Antagonist Protocols Versus Long GnRH Agonist Protocol in IVF/ICSI Cycles of Polycystic Ovary Syndrome Women: A Systematic Review and Meta-Analysis," *Scientific Reports* 12 (2022): 1–22.
8. S. Aggarwal and K. Pandey, "Early Identification of PCOS With Commonly Known Diseases: Obesity, Diabetes, High Blood Pressure and Heart Disease Using Machine Learning Techniques," *Expert Systems with Applications* 217 (2023): 119532.
9. E. Y. Adashi, D. Cibula, M. Peterson, and R. Azziz, "The Polycystic Ovary Syndrome: The First 150 Years of Study," *F&S Reports* 4, no. 1 (2023): 2–18.
10. K. S. Miles, *Our Pearls Matter: PCOS Through the Lens of Women of Color and White Women* (Ruston, LA: Louisiana Tech University, 2023).
11. S. A. Bhat, *Detection of Polycystic Ovary Syndrome Using Machine Learning Algorithms* (Dublin: National College of Ireland, 2021).
12. J. J. Benjamin, K. MaheshKumar, V. Radha, et al., "Stress and Polycystic Ovarian Syndrome-a Case Control Study Among Indian Women," *Clinical Epidemiology and Global Health* 22 (2023): 101326.
13. S. Karkera, E. Agard, and L. Sankova, "The Clinical Manifestations of Polycystic Ovary Syndrome (PCOS) and the Treatment Options," *European Journal of Biology and Medical Science Research* 11, no. 1 (2023): 57–91.
14. G. Usha Kiran, D. Yalla Vali, D. Yaswanth-Shankar, G. Lokesh, P. Krishna Chaitanya, and S. Sujitha, "Overview of Polycystic Ovary Syndrome (PCOS)," *World Journal of Advanced Engineering Technology and Sciences* 8, no. 2 (2023): 11–22.
15. S. F. M. Dadoush, "Diagnosing and Treating the Causes of women's Polycystic Ovary Syndrome: Clinical and Prospective Study," *African Journal of Advanced Pure and Applied Sciences* 2 (2023): 401–407.
16. A. Pramod, *Dietary and Physical Activity Pattern IN PCOS Women* (Ernakulam: St Teresa's College (Autonomous), 2023).
17. L. Della Corte, D. Boccia, M. Palumbo, et al., "Is There Still a Place for Surgery in Patients With PCOS? A Review," *Lifestyles* 13, no. 6 (2023): 1270.
18. M. M. Rahman, M. K. Nasir, M. Nur-A-Alam, and M. S. I. Khan, "Proposing a Hybrid Technique of Feature Fusion and Convolutional Neural Network for Melanoma Skin Cancer Detection," *Journal of Pathology Informatics* 14 (2023): 100341.
19. I. Silva, C. N. Ferreira, L. B. X. Costa, et al., "Polycystic Ovary Syndrome: Clinical and Laboratory Variables Related to New Phenotypes Using Machine-Learning Models," *Journal of Endocrinological Investigation* 45 (2022): 1–9.
20. Rahman, M.M., Islam A., Islam F., Zaman M., Islam M.R., Alam Sakib M.S., Hasan Babu H.M., "Empowering Early Detection: A Web-Based Machine Learning Approach for PCOS Prediction," *Informatics in Medicine Unlocked*, (London, UK: Elsevier, 2024). 47, 101500
21. V. V. Khanna, K. Chadaga, N. Sampathila, S. Prabhu, V. Bhandage, and G. K. Hegde, "A Distinctive Explainable Machine Learning Framework for Detection of Polycystic Ovary Syndrome," *Applied System Innovation* 6, no. 2 (2023): 32.
22. S. Bharati, P. Podder, and M. R. H. Mondal, "Diagnosis of Polycystic Ovary Syndrome Using Machine Learning Algorithms," in *2020 IEEE Region 10 Symposium (TENSYP)* (Piscataway, NJ: IEEE, 2020).
23. A. Zigarelli, Z. Jia, and H. Lee, "Machine-Aided Self-Diagnostic Prediction Models for Polycystic Ovary Syndrome: Observational Study," *JMIR Formative Research* 6, no. 3 (2022): e29967.
24. H. Danaei Mehr and H. Polat, "Diagnosis of Polycystic Ovary Syndrome Through Different Machine Learning and Feature Selection Techniques," *Health Technology* 12 (2022): 137–150.
25. Z. Zad, V. S. Jiang, A. T. Wolf, et al., "Predicting Polycystic Ovary Syndrome With Machine Learning Algorithms From Electronic Health Records," *Frontiers in Endocrinology* 15 (2024): 1298628.
26. S. Vedpathak, Pcos Dataset, 2020, <https://www.kaggle.com/shreyasvedpathak/pcos-dataset>.
27. A. G. Asuero, A. Sayago, and A. G. González, "The Correlation Coefficient: An Overview," *Critical Reviews in Analytical Chemistry* 36, no. 1 (2006): 41–59.
28. K. M. M. Uddin, M. J. Alam, M. A. Uddin, and S. Aryal, "A Novel Approach Utilizing Machine Learning for the Early Diagnosis of Alzheimer's Disease," *Biomedical Materials & Devices* 1 (2023): 1–17.
29. S. Maza and M. Touahria, "Feature Selection for Intrusion Detection Using New Multi-Objective Estimation of Distribution Algorithms," *Applied Intelligence* 49 (2019): 4237–4257.
30. J. Han, J. Pei, and M. Kamber, *Data Mining: Concepts and Techniques* (Amsterdam, Netherlands: Elsevier, 2011).
31. J. S. Madushani, et al., "Evaluating Expressway Traffic Crash Severity by Using Logistic Regression and Explainable & Supervised Machine Learning Classifiers," *Transportation Engineering* 13 (2023): 100190.
32. R. Praveena, et al., "Prediction of Rainfall Analysis Using Logistic Regression and Support Vector Machine," *Journal of Physics: Conference Series* 2466 (2023): 012032.
33. E. Najjar and A. M. Breesam, "Supervised Machine Learning a Brief Survey of Approaches," *Al-Iraqia Journal for Scientific Engineering Research* 2, no. 4 (2023): 71–82.
34. M. Çakir, M. Yilmaz, M. A. Oral, H. Ö. Kazanci, and O. Oral, "Accuracy Assessment of RFerns, NB, SVM, and kNN Machine Learning Classifiers in Aquaculture," *Journal of King Saud University, Science* 35, no. 6 (2023): 102754.
35. A. Tyagi, V. P. Singh, and M. M. Gore, "An Efficient Automated Detection of Schizophrenia Using k-NN and Bag of Words Features," *SN Computer Science* 4, no. 5 (2023): 518.
36. K. M. M. Uddin, R. Ripa, N. Yeasmin, N. Biswas, and S. K. Dey, "Machine Learning-Based Approach to the Diagnosis of Cardiovascular Vascular Disease Using a Combined Dataset," *Intelligence-Based Medicine* 7 (2023): 100100.
37. K. M. M. Uddin, R. Ripa, N. Yeasmin, N. Biswas, and S. K. Dey, "Machine Learning-Based Approach to the Diagnosis of Cardiovascular Vascular Disease Using a Combined Dataset," *Intelligence-Based Medicine* 7 (2023): 100100.
38. X. Meng, P. Zhang, Y. Xu, and H. Xie, "Construction of Decision Tree Based on C4. 5 Algorithm for Online Voltage Stability Assessment," *International Journal of Electrical Power & Energy Systems* 118 (2020): 105793.

39. Y. Ding, H. Zhu, R. Chen, and R. Li, "An Efficient AdaBoost Algorithm With the Multiple Thresholds Classification," *Applied Sciences* 12, no. 12 (2022): 5872.
40. A. A. Ibrahim, R. L. Ridwan, M. M. Muhammed, R. O. Abdulaziz, and G. A. Saheed, "Comparison of the CatBoost Classifier With Other Machine Learning Methods," *International Journal of Advanced Computer Science and Applications* 11, no. 11 (2020): 738–748, <https://doi.org/10.14569/IJACSA.2020.0111190>.
41. L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased Boosting With Categorical Features," *Advances in Neural Information Processing Systems* 31 (2018): 1–11.
42. M. Lu, Z. Fan, B. Xu, et al., "Using Machine Learning to Predict Ovarian Cancer," *International Journal of Medical Informatics* 141 (2020): 104195.
43. M. Akazawa and K. Hashimoto, "Artificial intelligence in ovarian cancer diagnosis," *Anticancer Research* 40, no. 8 (2020): 4795–4800.
44. J. Martinez-Mas, A. Bueno-Crespo, P. J. Garcia-Laencina, et al., *A Benign Ovarian Tumour Developing an Endometrial Cancer?* (London, UK: BioMed Central Ltd., 2015).
45. Y. Yuan, R. Wang, M. Luo, et al., "A Machine Learning Approach Using XGBoost Predicts Lung Metastasis in Patients With Ovarian Cancer," *BioMed Research International* 2022 (2022): 8501819.
46. N. B. Shannon, L. L. Y. Tan, Q. X. Tan, J. W. S. Tan, and J. Hendrikson, "A Machine Learning Approach to Identify Predictive Molecular Markers for Cisplatin Chemosensitivity Following Surgical Resection in Ovarian Cancer," *Scientific Reports* 11, no. 1 (2021): 16829.
47. M. Alam, J.-E. Anawar, K. M. Mohi Uddin, Md. H. Rahman, and Md. M. Rahman, "Machine Learning Techniques for Predicting Ovarian Cancer in its Early Stages Using Biomarkers," in *2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)* (Piscataway, NJ: Institute of Electrical and Electronics Engineers (IEEE), 2024), 1257–1262.
48. M. Ahmed, H. Rehana, and S. Asaduzzaman, "Ovarian Cancer Substantial Risk Factor Analysis by Machine Learning: A Low Incoming Country Perspective," *Biointerface Research in Applied Chemistry* 11 (2021): 8457–8466.
49. E. Kawakami, J. Tabata, N. Yanaihara, et al., "Application of Artificial Intelligence for Preoperative Diagnostic and Prognostic Prediction in Epithelial Ovarian Cancer Based on Blood Biomarkers," *Clinical Cancer Research* 25, no. 10 (2019): 3006–3015.
50. S. Suha and M. Islam, "Exploring the Dominant Features and Data-Driven Detection of Polycystic Ovary Syndrome Through Modified Stacking Ensemble Machine Learning Technique," *Heliyon* 9, no. 3 (2023): e14518.